

Chapter 109

Matte Nodes

This chapter details the Matte nodes available in Fusion.

The abbreviations next to each node name can be used in the Select Tool dialog when searching for tools and in scripting references.

For purposes of this document, node trees showing MediaIn nodes in DaVinci Resolve are interchangeable with Loader nodes in Fusion Studio, unless otherwise noted.

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Alpha Divide [ADv]



The Alpha Divide node

Alpha Divide Node Introduction

As the name gives away, the Alpha Divide's sole purpose is to divide an incoming image's color channels by its alpha channel. When you color correct an image that contains a premultiplied alpha channel, first apply an Alpha Divide node before any color correction node to create a non-premultiplied image. Then you can perform the color correction. After the color correction, add an Alpha Multiply node to return the image to its premultiplied state.

Inputs

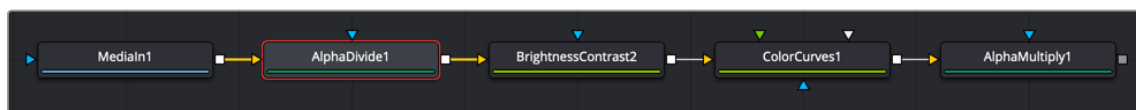
The Alpha Divide node includes two inputs in the Node Editor.

Input: The orange input accepts a 2D image with a premultiplied Alpha.

Effect Mask: The optional blue input expects a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps masks. Connecting a mask to this input limits the pixels where the Alpha divide occurs. An effects mask is applied to the tool after the tool is processed.

Basic Node Setup

The Alpha Divide node is placed before any color correction is done to an image with a premultiplied Alpha channel. Here the MediaIn node is assumed to have a premultiplied Alpha channel. The Alpha Divide node is inserted, and then color correction nodes operate on the "straight" Alpha. An Alpha Multiply node is placed at the end of the chain to premultiply the Alpha channel again. If only a single color correction node is used, then the Pre-Divide/Post-Multiply checkbox on the Options tab can be used in place of the Alpha Divide/Alpha Multiple nodes.



An Alpha Divide node is inserted before color correcting an image with premultiplied alpha.

Inspector

This node has no controls.

Alpha Multiply [AMI]



The Alpha Multiply node

Alpha Multiply Node Introduction

As the name gives away, the Alpha Multiply's sole purpose is to multiply an image's color channels by its alpha channel. When you color correct an image that contains a premultiplied alpha channel, first apply an Alpha Divide node before any color correction node to create a non-premultiplied image. Then you can perform the color correction. After the color correction, add an Alpha Multiply node to return the image to its premultiplied state.

Inputs

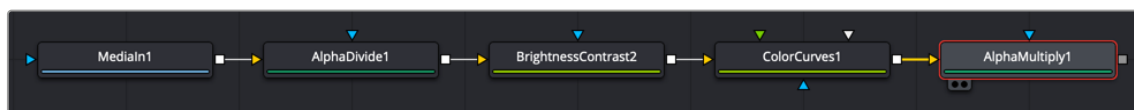
The Alpha Multiply node includes two inputs in the Node Editor.

Input: The orange input accepts a 2D image with a "straight" or non-premultiplied alpha.

Effect Mask: The optional blue input expects a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps masks. Connecting a mask to this input limits the pixels where the Alpha multiply occurs. An effects mask is applied to the tool after the tool is processed.

Basic Node Setup

The Alpha Multiply node is placed after color correction is done to an image with a premultiplied Alpha channel. Here the MediaIn node is assumed to have a premultiplied Alpha channel. The Alpha Divide node is inserted, and then color correction nodes operate on the "straight" Alpha. An Alpha Multiply node is placed at the end of the chain to premultiply the Alpha channel again. If only a single color correction node is used, then the Pre-Divide/Post-Multiply checkbox on the Options tab can be used in place of the Alpha Divide/Alpha Multiple nodes.

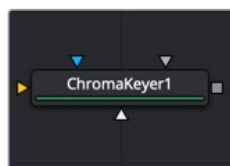


An Alpha Multiply node is inserted after color correcting an image with premultiplied alpha.

Inspector

This node has no controls.

Chroma Keyer [CKy]



The Chroma Keyer node

Chroma Keyer Node Introduction

The Chroma Keyer node creates an alpha channel (matte) for an image by removing selected colors from the scene. Unlike the Delta Keyer or Primatte, which use specific optimizations for keying from blue and green colors, the Chroma Keyer works equally well with any color.

NOTE: When working with blue- or green-screen shots, it is best to use the Delta Keyer or Primatte node, rather than the more general purpose Chroma Keyer node.

Inputs

The Chroma Keyer node includes four inputs in the Node Editor.

Input: The orange input accepts a 2D image that contains the color you want to be keyed for transparency.

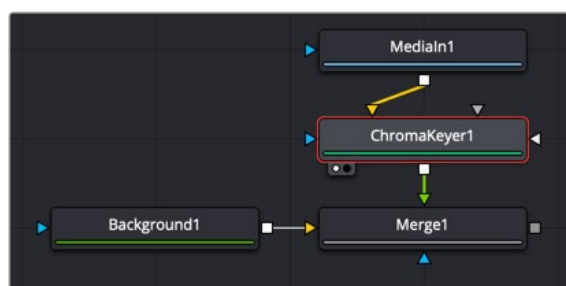
Garbage Matte: The gray garbage matte input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps masks. Connecting a mask to this input causes areas of the image that fall within the matte to be made transparent. The garbage matte is applied directly to the alpha channel of the image.

Solid Matte: The white solid matte input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps masks. Connecting a mask to this input causes areas of the image that fall within the matte to be fully opaque.

Effect Mask: The optional blue input expects a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps masks. Connecting a mask to this input limits the pixels where the alpha multiply occurs. An effects mask is applied to the tool after the tool is processed.

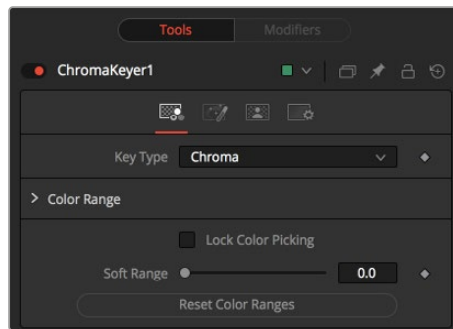
Basic Node Setup

The Chroma Keyer node takes in a 2D image like the MediaIn node below and removes a color that you identify by dragging over it in the viewer. The result is that the selected color is replaced with transparency, allowing you to composite the image as the foreground in a Merge node.



A Chroma Keyer node creating transparency on the MediaIn node.

Inspector



The Chroma Key tab

Chroma Key Tab

The Chroma Key tab is used to make the initial selection of color for keying.

Key Type

The Key Type menu determines the selection method used for the matte creation.

- **Chroma:** The Chroma method creates a matte based on the RGB values of the selected color range.
- **Color:** The Color method creates a matte based on the hue of the selected color range.

Color Range

Colors are made transparent by selecting the Chroma Keyer node in the node tree, and then dragging a selection around the colors in the viewer. The range controls update automatically to represent the current color selection. You can tweak the range sliders slightly, although most often selecting colors in the displays is all that is required.

Lock Color Picking

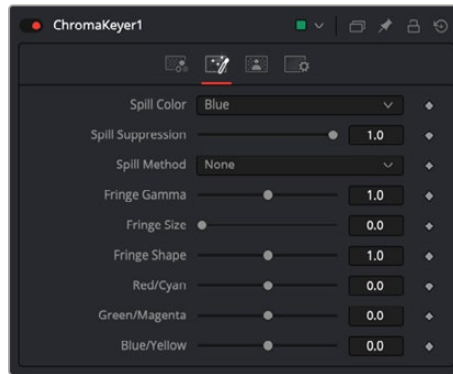
When this checkbox is activated, selecting colors from the viewer is disabled to prevent accidental addition to the range. It is a good idea to activate this checkbox once you make the color selection for the matte. All other controls in the node remain editable.

Soft Range

This control softens the selected color range, adding additional colors into the matte.

Reset Color Ranges

Clicking this button resets the Chroma Keyer's range controls, discarding all color selections. All other sliders and controls maintain their values.



The Chroma Key Image tab

Image Tab

The Image tab primarily handles removing spill color on the foreground subject. Color spill occurs when light containing the color you are removing is reflected onto the foreground subject.

Spill Color

This menu selects the color used as the base for all spill suppression techniques.

Spill Suppression

This slider sets the amount of spill suppression applied to the foreground subject.

When this slider is set to 0, no spill suppression is applied.

Spill Method

This menu selects the strength of the algorithm used to apply spill suppression to the image.

- **None:** None is selected when no spill suppression is required.
- **Rare:** This removes very little of the spill color and is the lightest of all methods.
- **Medium:** This works best for green screens.
- **Well Done:** This works best for blue screens.
- **Burnt:** This works best for blue. Use this mode only for very troublesome shots. Most likely you will have to add strong color correction after the key to get, for example, your skin tones back.

Fringe Gamma

This control is used to adjust the brightness of the fringe or halo that surrounds the keyed image.

Fringe Size

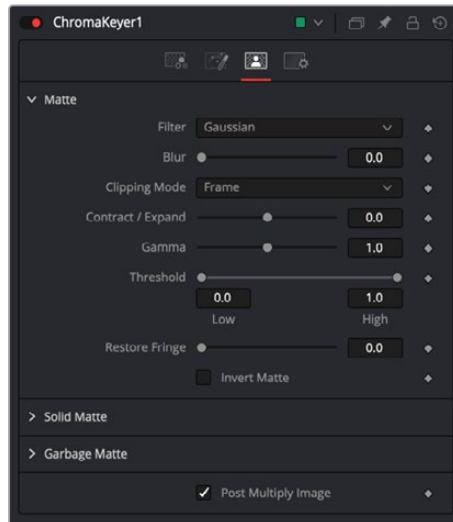
This expands and contracts the size of the fringe or halo surrounding the keyed image.

Fringe Shape

Fringe Shape forces the fringe toward the external edge of the image or toward the inner edge of the fringe. Its effect is most noticeable while the Fringe Size slider's value is large.

Cyan/Red, Magenta/Green, and Yellow/Blue

Use these three controls to color correct the fringe of the image. This is useful for correcting semitransparent pixels that still contain color from the original background to match the new background.



The Chroma Key Matte tab

Matte Tab

The Matte tab refines the softness, density, and overall fit of the resulting matte.

Filter

This control selects the filtering algorithm used when applying blur to the matte.

- **Box:** This is the fastest method but at reduced quality. Box is best suited for minimal amounts of blur.
- **Bartlett:** Otherwise known as a Pyramid filter, Bartlett makes a good compromise between speed and quality.
- **Multi-box:** When selecting this filter, the Num Passes slider appears and lets you control the quality. At 1 and 2 passes, results are identical to Box and Bartlett, respectively. At 4 passes and above, results are usually as good as Gaussian, in less time and with no edge “ringing.”
- **Gaussian:** The Gaussian filter uses a true Gaussian approximation and gives excellent results, but it is a little slower than the other filters. In some cases, it can produce an extremely slight edge “ringing” on floating-point pixels.

Blur

Matte Blur blurs the edge of the matte based on the Filter menu setting. A value of zero results in a sharp, cutout-like hard edge. The higher the value, the more blur applied to the matte.

Clipping Mode

This option determines how edges are handled when performing domain of definition rendering. This is profoundly important when blurring the matte, which may require samples from portions of the image outside the current domain.

- **Frame:** The default option is Frame, which automatically sets the node’s domain of definition to use the full frame of the image, effectively ignoring the current domain of definition. If the upstream DoD is smaller than the frame, the remaining area in the frame will be treated as black/transparent.
- **Domain:** Setting this option to Domain will respect the upstream domain of definition when applying the node’s effect. This can have adverse clipping effects in situations where the node employs a large filter.

- **None:** Setting this option to None will not perform any source image clipping at all. This means that any data required to process the node's effect that would normally be outside the upstream DoD will be treated as black/transparent.

Contract/Expand

This slider shrinks or grows the semitransparent areas of the matte. Values above 0.0 expand the matte, while values below 0.0 contract it.

This control is usually used in conjunction with the Matte Blur to take the hard edge of a matte and reduce fringing. Since this control affects only semitransparent areas, it will have no effect on a matte's hard edge.

Gamma

Matte Gamma raises or lowers the values of the matte in the semitransparent areas. Higher values cause the gray areas to become more opaque, and lower values cause the gray areas to become more transparent. Completely black or white regions of the matte remain unaffected.

Since this control affects only semitransparent areas, it will have no effect on a matte's hard edge.

Threshold

This range slider sets the lower threshold using the handle on the left and sets the upper threshold using the handle on the right.

Any value below the lower threshold setting becomes black or transparent in the matte.

Any value above the upper threshold setting becomes white or opaque in the matte. All values within the range maintain their relative transparency values.

This control is often used to reject salt and pepper noise in the matte.

Restore Fringe

This restores the edge of the matte around the keyed subject. Often when keying, the edge of the subject where you have hair is clipped out. Restore Fringe brings back that edge while keeping the matte solid.

Invert Matte

When this checkbox is selected, the alpha channel created by the keyer is inverted, causing all transparent areas to be opaque and all opaque areas to be transparent.

Solid Matte

Solid Mattes are mask nodes or images connected to the solid matte input on the node. The solid matte is applied directly to the alpha channel of the image. Generally, solid mattes are used to hold out keying in areas you want to remain opaque, such as someone with blue eyes against a blue screen.

Enabling Invert will invert the solid matte, before it is combined with the source alpha.

Garbage Matte

Garbage mattes are mask nodes or images connected to the garbage matte input on the node. The garbage matte is applied directly to the alpha channel of the image. Generally, garbage mattes are used to remove unwanted elements that cannot be keyed, such as microphones and booms. They are also used to fill in areas that contain the color being keyed but that you wish to maintain.

Garbage mattes of different modes cannot be mixed within a single tool. A Matte Control node is often used after a Keyer node to add a garbage matte with the opposite effect of the matte applied to the keyer.

Enabling Invert will invert the garbage matte, before it is combined with the source alpha.

Post-Multiply Image

Select this option to cause the keyer to multiply the color channels of the image against the alpha channel it creates for the image. This option is usually enabled and is on by default.

Deselect this checkbox and the image can no longer be considered premultiplied for purposes of merging it with other images. Use the Subtractive option of the Merge node instead of the Additive option.

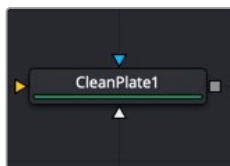
For more information on these Merge node settings, see *Chapter 95, "Composite Nodes,"* in the DaVinci Resolve Reference Manual, or Chapter 35 in the Fusion Reference Manual.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other matte nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Clean Plate



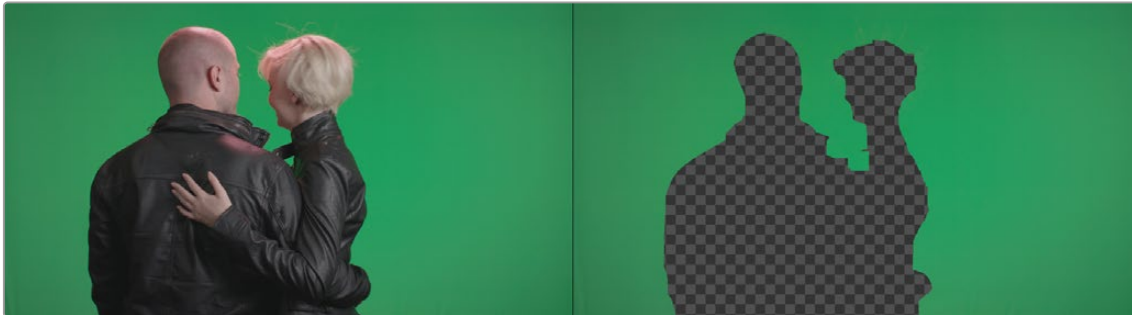
The Clean Plate node

Clean Plate Node Introduction

The Clean Plate tool is a pre-keying node used to generate an image of the green or blue color screen to smooth out the lighting differences. The output of the Clean Plate is later connected to the Clean Plate input on the Delta Keyer so it can key fine detail without choking or clipping the matte.

How to Create a Clean Plate

Creating a clean plate is the opposite of creating a key. When keying, you try to remove the green or blue color. When creating a clean plate, you try to keep as much of the blue- or green-screen as possible. By box selecting areas of the screen color in the viewer, you end up with an image of the green/blue screen. A transparent cutout represents everything that is not part of the blue or green screen.



Once you have the selection, the Erode control expands the pre-matte, removing any small pixels of non-green/blue screen around the edges. Then, growing the pre-matte fills in the holes until you have a solid blue or green image.

Inputs

The Clean Plate node includes three inputs in the Node Editor.

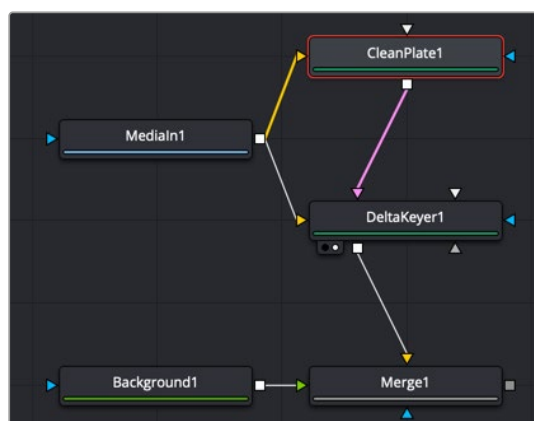
Input: The orange input accepts a 2D image that contains the green or blue screen.

Garbage Matte: The white garbage matte input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps masks. Connecting a mask to this input causes areas of the image that fall within the matte to be excluded from the clean plate. For a clean plate, garbage mattes should contain areas that are not part of the blue or green screen.

Effect Mask: The optional blue input expects a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps masks. Connecting a mask to this input limits the pixels where the clean plate is generated. An effects mask is applied to the tool after the tool is processed.

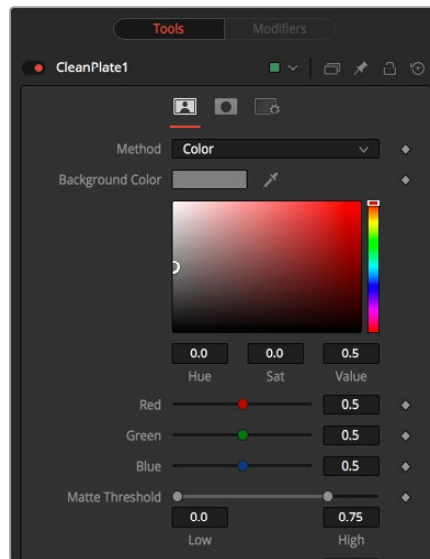
Basic Node Setup

The Clean Plate node and the Delta Keyer are two separate branches stemming from the main image you want to key. The green-screen or blue-screen clip is breached to connect to both the orange image input on the Clean Plate and the orange image input on the Delta Keyer. The output of the clean plate is then connected to the magenta clean plate input on the Delta Keyer. The output of the Delta Keyer is then used as the foreground to a Merge.



A Clean Plate connected to a Delta Keyer

Inspector



The Clean Plate tab

Plate Tab

The Plate tab contains the primary tools for creating a clean plate. Using this tab, you drag over the areas in the viewer, and then use the Erode and Grow Edges sliders to create the clean plate.

Method

The Method menu selects the type of color selection you use when sampling colors in the viewer.

- **Color:** Color uses a difference method to separate the background color. This works well on screen colors that are even.
- **Ranges:** Ranges uses a chroma range method to separate the background color. This is a better option for shadowed screen or screens that have different colors.

Matte Threshold

This range slider sets the lower threshold using the handle on the left and sets the upper threshold using the handle on the right.

Any value below the lower threshold becomes black or transparent in the matte.

Any value above the upper threshold becomes white or opaque in the matte. All values within the range maintain their relative transparency values. This control is often used to reject salt and pepper noise in the matte.

Erode

The Erode slider decreases the size of the screen area. It is used to eat away at small non-screen color pixels that may interfere with creating a smooth green- or blue-screen clean plate.

Crop

Crop trims in from the edges of the image.

Grow Edges

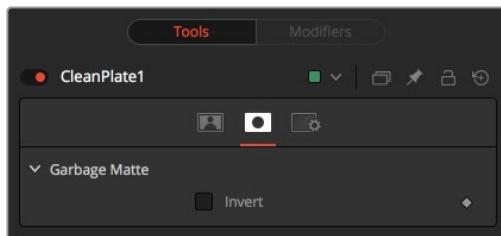
The Grow Edges slider expands the color of the edges to fill in holes until fully green or blue screen is created.

Fill

The Fill checkbox fills in remaining holes with color from the surrounding screen color.

Time Mode

- **Sequence:** Generates a new clean plate every frame.
- **Hold Frame:** Holds the clean plate at a single frame.



The Clean Plate Mask tab

Mask Tab

The Mask tab is used to invert the mask connected to the garbage mask input on the node. The garbage mask can be applied to clear areas before growing edges or filling remaining holes.

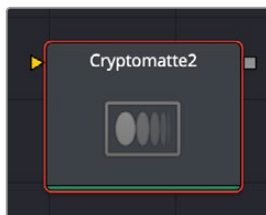
Invert

Invert uses the transparent parts of the mask to clear the image.

Common Controls**Settings Tab**

The Settings tab in the Inspector is also duplicated in other matte nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Cryptomatte [Cry] (Studio Version Only)



The Cryptomatte node

Cryptomatte Node Introduction

The Cryptomatte tool in Fusion allows you to read embedded mattes rendered into an EXR. Instead of relying on manually-assigned masks or color keying, Cryptomatte reads encoded IDs and isolates them in a way that allows for easy selection. This tool allows you to create mattes for specific objects or materials in a 3D-rendered scene.

Inputs

The Cryptomatte node has only one input in the Node Editor.

Input: The orange input accepts a 2D EXR image with embedded mattes.

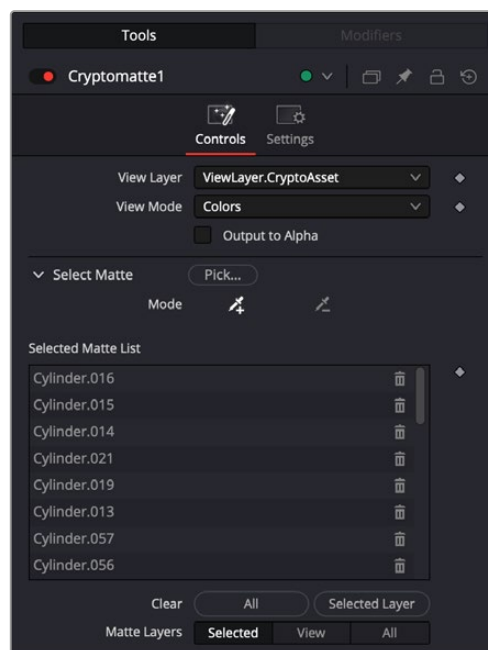
Basic Node Setup

The Cryptomatte node takes an EXR image with embedded mattes, then lets you select which mattes to pass through to the rest of the comp.



Cryptomatte taking in an EXR, selecting its mattes, and then passing them on to the Color Corrector and Bitmap tools

Inspector



The Cryptomatte Controls

These parameters let you control the matte selection of an EXR file.

- **View Layer:** Change the layer and matte type to pick (Object, Material etc). This affects the type of matte you can choose with the Pick control.
- **View Mode:** Changes the output of the Cryptomatte tool.

Colors: Shows the layer IDs as separate colors.

Edges: Shows the source EXR with outlined mattes overlaid the image.

Beauty: Shows the source EXR.

Matte: Outputs a black and white image of the selected matte.

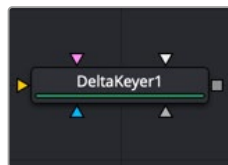
- **Select Matte:** Select mattes from the viewer or from a list.
Pick: Click to open a pick window with a matte list selection. Click and drag or use Cmd/Control-clicks to select multiple mattes from the list.
Mode: Click and drag the control to get an eye dropper cursor to select or remove mattes from the viewer.
- **Selected Matte List:** The Selected Matte List displays the selected mattes in a list view. The selection is highlighted in yellow on the viewer.
Clear: Click the bin icon to remove the selected mattes.
Clear All: Click this button to remove all mattes from the Selected Matte List.
Clear Selected Layer: Click to remove all mattes from the layer.
Matte Layers: Toggle between only the Selected layers, View, and All.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other matte nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Delta Keyer



The Delta Keyer node

Delta Keyer Node Introduction

The Delta Keyer is an advanced color difference keyer, with many features and controls for tuning the matte and separating the subject from a blue or green screen.

It contains several keying systems: the Key tab is the master difference keyer and Pre-Matte is a built-in clean plate to smooth out screen color. Tuning, Fringe, and Matte finish the keying process. The tabs are laid out in the rough order you tend to use them.

Inputs

The Delta Keyer node includes five inputs in the Node Editor.

Input: The orange input accepts a 2D image that contains the color you want to be keyed for transparency.

Garbage Matte: The gray garbage matte input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps masks. Connecting a mask to this input causes areas of the image that fall within the matte to be made transparent. The garbage matte is applied directly to the alpha channel of the image.

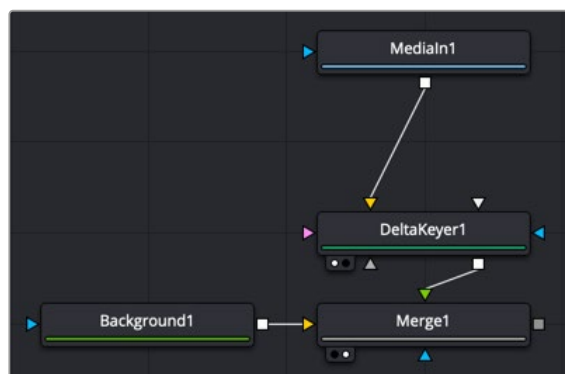
Solid Matte: The white solid matte input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps masks. Connecting a mask to this input causes areas of the image that fall within the matte to be fully opaque.

Clean Plate: Accepts the resulting image from the Clean Plate node.

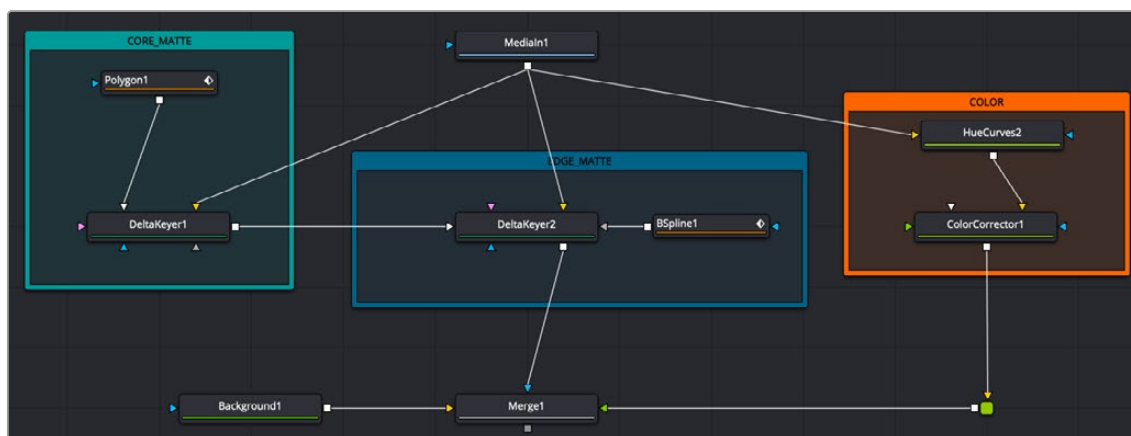
Effect Mask: The optional blue input expects a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps masks. Connecting a mask to this input limits the pixels where the keying occurs. An effects mask is applied to the tool after the tool is processed.

Basic Node Setup

The first operation needed to use the Delta Keyer is to open the Delta Keyer in the Inspector, then click and drag the background color eyedropper tool into the Viewer to select the color you want to key out. This color will then turn gray in the Viewer. A single Delta Keyer rarely gets perfect results because most green- or blue-screen shots have problems that the keyer is not made to handle. Keyers often need the help of garbage mattes or solid mattes created with a Polygon or B-Spline node. Shots can also require more than just one keyer to achieve perfect results. That is shown in one of the examples below, where one Delta Keyer is used to create a hard inner white matte, while a second Delta Keyer is used to capture the soft edges. The core matte below is then connected into the solid matte input on the second Delta Keyer. Color correction, including spill suppression, is best handled as a separate branch in your node tree. Separating your matte from your color correction using underlays makes it clear what branch of nodes is performing what operation.

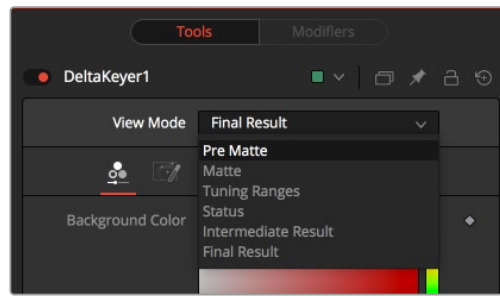


Starting point for a Delta Keyer setup



An expanded key setup with two Delta Keyers, a Polygon and B-Spline nodes for mattes, and a branch for color correction

Inspector

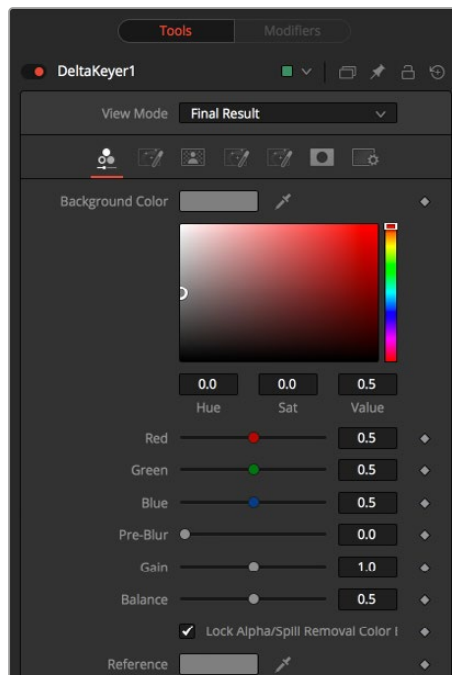


The Delta Keyer View Mode menu

Key Tab View Mode

At the top of the Inspector is the View Mode menu. The default selection shows the final result. You can change the view to see various intermediate stages of the keying process.

- **Pre Matte:** Displays the output of the Pre Matte key.
- **Matte:** Displays the alpha of the key before being combined with solid and garbage masks. When displaying the matte, set the viewer to show the alpha channel.
- **Tuning Ranges:** Displays a false color for Shadow, Midtone, and Highlight ranges of the image. Shadows are in the red channel, Midtones are in the green channel, and Highlights are in the blue channel.
- **Status:** Displays information to indicate areas that are solid, transparent, or in between. It also displays areas that have been affected by matte adjustments, such as thresholding or erode/dilate and areas affected by the solid mask.
- **Intermediate Result:** The untouched source image color channels combined with the final matte. Optionally, combine this output with additional Delta Keyer nodes.
- **Final Result:** The final keyed image with spill suppression, ready to merge onto a scene.



The Delta Keyer Key tab

Key Tab

The Key tab is where most keying begins. It is used to select the screen color.

Background Color

This is the color of the blue or green screen, sometimes called the screen color. To create the key with the Delta Keyer, use the background color Eyedropper to select the screen color from the image.

Pre-Blur

Applies a blur before generating the alpha. This can help with certain types of noise, edge enhancements, and artifacts in the source image.

Gain

Gain increases the influence of the screen color, causing those areas to become more transparent.

Balance

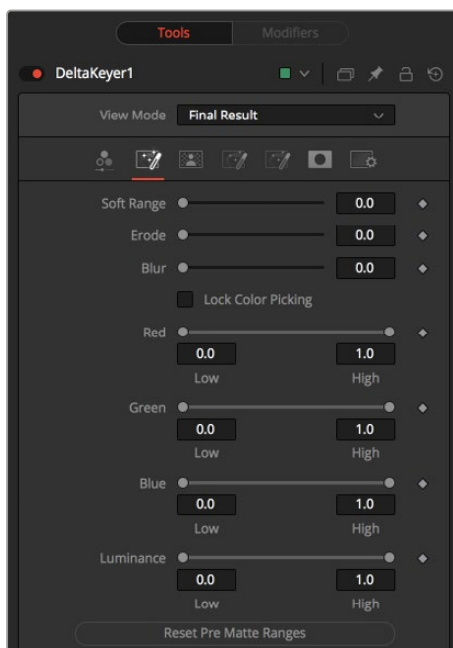
A color difference keyer, like the Delta Keyer, compares the differences between the dominant channel determined by the selected background color and the other two channels. Adjusting balance determines the proportions of the other two channels. A value of 0 uses the minimum of the other two channels, where a value of 1 uses the maximum. A value of 0.5 uses half of each.

Lock Alpha/Spill Removal Color Balance Reference

Unlocking this allows you to use different color references when generating the alpha and when determining how much of the background color to subtract from the image. When enabled, spill and color are combined.

Color Balance Reference

The Color balance reference controls adjust for lighting or white balance that might be reducing background color purity and saturation. A correction is applied based on the reference of a neutral-colored object when generating the key and determining the amount of background color subtraction, without altering the background color that is subtracted.



The Delta Keyer Pre Matte tab

Pre Matte Tab

The Pre Matte tab is in some ways a garbage matte keyer that occurs before the delta keyer process. It is used to even out the screen color before a more delicate key is done. After selecting the initial screen color, you can switch to the Pre Matte tab and view the PreMatte Out from view mode menu at the top of the Inspector. The Pre Matte works on ranges of color, so you start by dragging a bounding box around areas where lighting is different from the initial screen color pick. Use the Soft range to expand, and Erode to cut away the edges.

Once you have a more even screen selection, you can move to the Matte tab.

Soft Range

The Soft Range extends the range of selected color and rolloff of the screen color.

Erode

Erode contracts the edge of the pre matte, so the edge detail does not clip.

Blur

This softens the edges of the pre matte.

Pre Matte Range

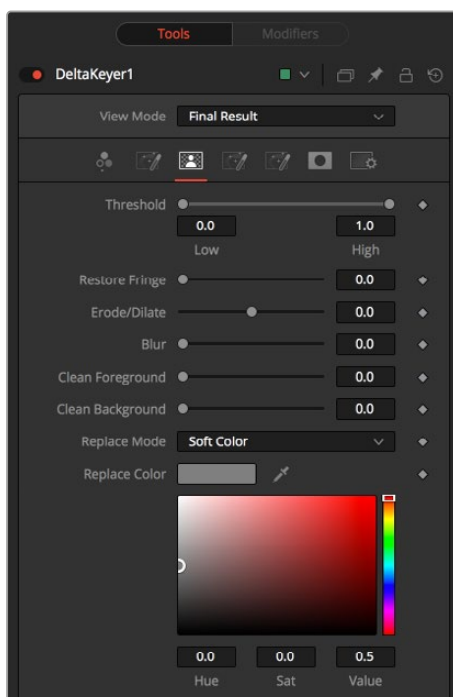
These controls update automatically to represent the current color selection. The controls tweak the selection slightly, although selecting colors in the viewer is all that is required.

Lock Color Picking

When this checkbox is activated, it prevents the accidental growth of the selected range by selecting more colors from the view. It is a good idea to activate this checkbox after selecting the color for the matte. All other controls in the node remain editable.

Reset Pre Matte Ranges

This button discards all color selection by resetting the ranges but maintains all other slider and control values.



The Delta Keyer Matte tab

Matte Tab

The Matte tab refines the alpha of the key, combined with any solid and garbage masks connected to the node. When using the matte tab, set the viewer to display the alpha channel of the Delta Keyer's final output.

Threshold

This range slider sets the lower threshold using the handle on the left and sets the upper threshold using the handle on the right.

Any value below the lower threshold setting becomes black or transparent in the matte.

Any value above the upper threshold setting becomes white or opaque in the matte. All values within the range maintain their relative transparency values.

Restore Fringe

This restores the edge of the matte around the keyed subject. Often when keying, clipping occurs around the edge of the subject where there is hair. Restore Fringe brings back that edge while keeping the matte solid.

Erode/Dilate

Expands or contracts the matte.

Blur

Softens the matte.

Clean Foreground

Fills slightly transparent (light gray) areas of the matte.

Clean Background

Clips the bottom dark range of the matte.

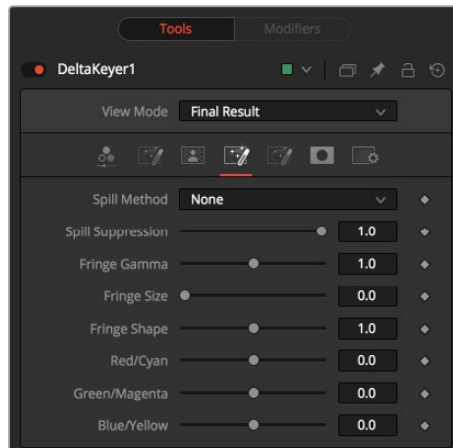
Replace Mode

Determines how matte adjustments restore color to the image.

- **None:** No color replacement. Matte processing does not affect the color.
- **Source:** The color from the original image.
- **Hard Color:** A solid color.
- **Soft Color:** A solid color weighted by how much background color was originally removed.

Replace Color

The color used with the Hard Color and Soft Color replace modes.



The Delta Keyer Fringe tab

Fringe Tab

The Fringe tab handles the majority of spill suppression in the Delta Keyer. Spill suppression is a form of color correction that attempts to remove the screen color from the fringe of the matte.

Spill is the transmission of the screen color through the semitransparent areas of the alpha channel. In the case of blue- or green-screen keying, this usually causes the color of the background to become apparent in the edges of the foreground subject.

Spill Method

This selects the strength of the algorithm used to apply spill suppression to the image.

- **None:** None is selected when no spill suppression is required.
- **Rare:** This removes very little of the spill color and is the lightest of all methods.
- **Medium:** This works best for green screens.
- **Well Done:** This works best for blue screens.
- **Burnt:** This works best for blue screens. Use this mode only for very troublesome shots.

Spill Suppression

When this slider is set to 0, no spill suppression is applied to the image. Increasing the slider increases the strength of the spill method.

Fringe Gamma

This control can be used to adjust the brightness of the fringe or halo that surrounds the keyed image.

Fringe Size

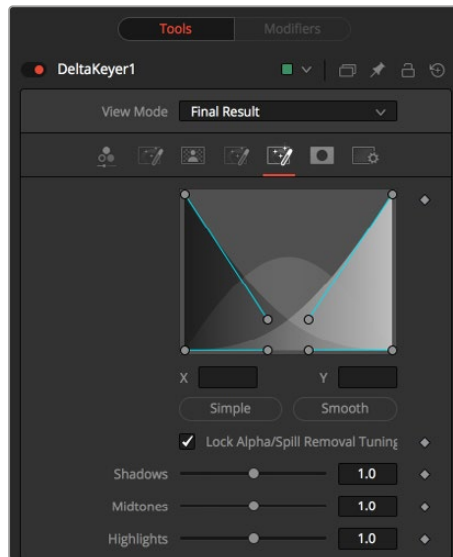
This expands and contracts the size of the fringe or halo surrounding the keyed image.

Fringe Shape

Fringe Shape presses the fringe toward the external edge of the image or pulls it toward the inner edge of the fringe. Its effect is most noticeable while the Fringe Size value is large.

Cyan/Red, Magenta/Green, and Yellow/Blue

Use these three controls to color correct the fringe of the image. This is useful for correcting semitransparent pixels that still contain color from the original background to match the new background.



The Delta Keyer Tuning tab

Tuning Tab

The Tuning tab is an advanced tab that allows you to determine the size of the shadow, midtone, and highlight ranges. By modifying the ranges, you can select the strength of the matte and spill suppression based on tonal values.

Range Controls

The range controls define how much color is considered to fall into the shadows, midtones, and highlights areas of the image. The spline controls allow for easy adjusting of the tonal ranges of each Shadow and Highlight tonal map.

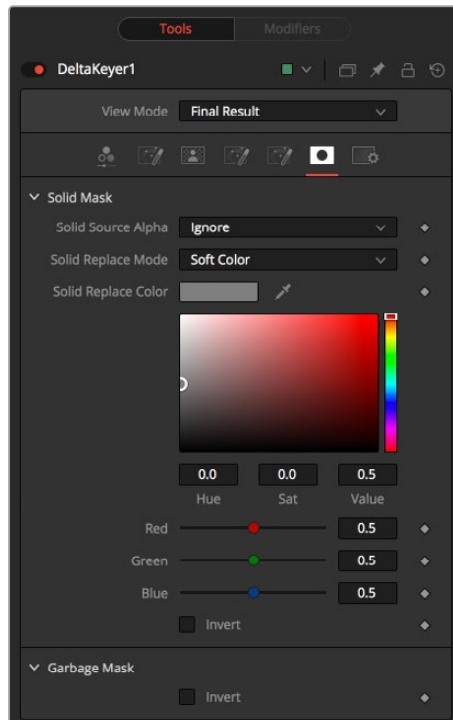
Simple/Smooth

The Simple button sets the range to be linear. The Smooth button sets a smooth tonal gradient for the ranges.

Lock Alpha/Spill Removal Tuning

When this checkbox is disabled, separate tuning controls are used when generating the alpha and when determining how much of the background color to subtract from the image.

- **Shadows:** Adjusts the strength of the key in darker areas of the background.
- **Midtones:** Adjusts the strength of the key in midtone areas of the background.
- **Highlights:** Adjusts the strength of the key in brighter areas of the background.



The Delta Keyer Mask tab

Mask Tab

The Mask tab determines how the solid and garbage mattes are applied to the key.

Solid Source Alpha

Used to combine the existing alpha from the source image with the solid mask.

- **Ignore:** Does not combine the alpha from the source image.
- **Add:** Solid areas of the source image alpha are made solid in the solid mask.
- **Subtract:** Transparent areas of the source image alpha are made transparent in the solid mask.

Solid Replace Mode

This determines how the solid mask restores color to the image.

- **None:** No color replacement. The solid mask does not affect the color.
- **Source:** The color from the original image.
- **Hard Color:** A solid color.
- **Soft Color:** A solid color weighted by how much background color was originally removed.

Solid Replace Color

The color used with the Hard Color and Soft Color replace modes.

- **Invert:** Inverts the solid mask, before it is combined with the source alpha.

Garbage Mask

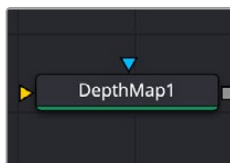
- **Invert:** Normally, solid areas of a garbage mask remove the image. When inverted, the transparent areas of the mask remove the image.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other matte nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Depth Map [DMp]



The Depth Map node

Depth Map Node Introduction

Depth Map creates an Alpha channel based on the perceived distance of objects in your clip. By being able to isolate a specific depth region, the opportunities to manipulate the resulting image are greatly expanded. For example, combined with a Blur node you could simulate a depth of field effect, but confined only to the distant background of your shot. You could enhance a foreground object, like a person, increasing contrast, saturation, and sharpness, similar to using a qualifier. You can use a Depth Map to fix video issues, for example fixing a color temperature problem by isolating a far window that was shot in daylight, causing background actors to be tinted blue, while leaving the foreground subject that was shot with studio lighting unaffected.

The resulting Depth Map Alpha channel is visualized as a black and white image, with white being the area that is affected by the resulting changes and black areas remaining unchanged.

Inputs

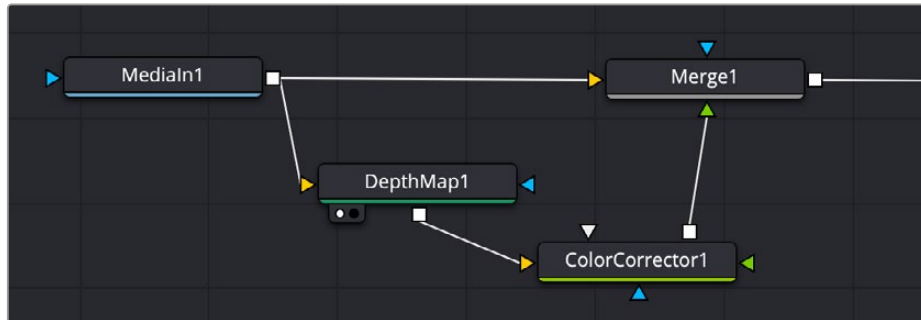
The Depth Map node includes two inputs in the Node Editor.

Input: The yellow input accepts a 2D image that contains the shot you wish to analyze for depth.

Effect Mask: The optional blue input expects a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps masks. Connecting a mask to this input limits the pixels where the difference matte occurs. An effects mask is applied to the tool after the tool is processed.

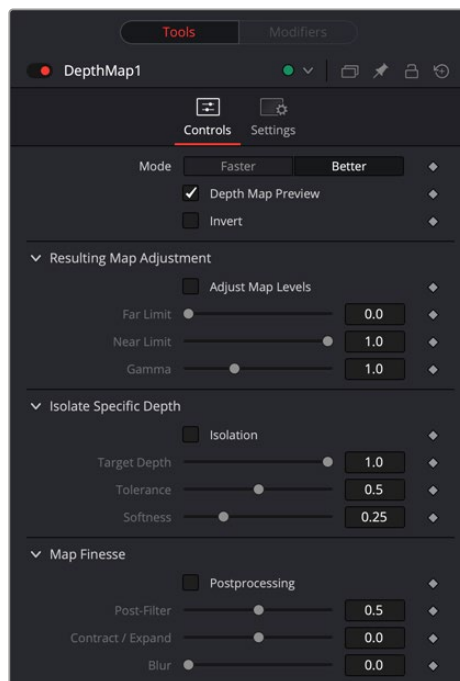
Basic Node Setup

The Depth Map is designed to take an input from an image source and create an Alpha channel based on the perceived distance from the camera. Then it will feed that Alpha to downstream nodes. In the example below, the Depth Map is used to isolate the background of the image and apply a color grade to just the background, while leaving the foreground actors unaffected as they move through the frame.



A sample Depth Map node structure for isolating subjects based on distance from the camera

Inspector



The Depth Map Controls tab

Controls Tab

The Controls tab in the Depth Map contains all the parameters for adjusting the quality of the resulting matte.

- **Mode:** Depth Map is a very computationally intensive effect. The quality setting allows a Faster mode to speed up the responsiveness to adjustments, while the default Better mode gives the best results and should be turned on when the adjustments are finished.

- **Depth Map Preview:** By default this box is checked and shows you the current Depth Map for making adjustments. When this check box is disabled, the resulting Alpha can then be used for grading on other nodes.
- **Invert:** Checking this box reverses the Depth Map, switching its transparent and opaque regions.

Resulting Map Adjustment

These controls let you determine how the Depth Map's contrast is adjusted.

- **Adjust Map Levels:** When deselected (default), all scaling is turned off, allowing you to adjust the full range of the Depth Map. When enabled, this option clips the Depth Map's levels to 0 and 1. This functions as a preview of what will happen to the Depth Map when used as an Alpha channel where the values are always clipped to 0 and 1. Checking this box also activates the tools below.
- **Far Limit:** This control adjusts the black levels of the Depth Map.
- **Near Limit:** This control adjusts the white levels of the Depth Map.
- **Gamma:** This control adjusts the intermediate depth values to be brighter or dimmer compared to the fixed black and white levels.

Isolate Specific Depth

These controls allow you to sweep backwards and forwards in the scene by depth, allowing you to isolate a specific depth range to adjust.

- **Isolation:** This checkbox turns the depth isolation tools on or off.
- **Target Depth:** This controls the specific depth you want to isolate. 1 is fully in the foreground, while 0 is fully in the background.
- **Tolerance:** Sets the range to either side of the Target Depth to include in the Depth Map.
- **Softness:** This sets a subtle ramp-in and ramp-out of the selected range, giving it a more organic selection.

Map Finesse

These controls modify the resulting Depth Map's Alpha channel for use in grading.

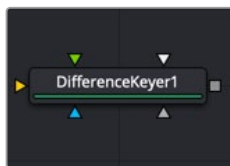
- **Post Processing:** This control turns the Map Finesse tools on or off.
- **Post-Filter:** This control blends the map to the smooth areas and edges of the image. It is used to prevent later grading effects from visibly varying within the region.
- **Contract/Expand:** This control dilates or erodes the overall shape at the edges; useful for fine tuning the boundary between the affected and unaffected regions of the map.
- **Blur:** This control softens the boundary of the map, allowing it to blend more smoothly into the resulting image.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Matte nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

Difference Keyer [DfK]



The Difference Keyer node

Difference Keyer Node Introduction

Difference keying is a process that produces a matte based on the differences between two images. A difference key uses two input images: one containing the subject with the background and another containing the background without the subject.

Although the process sounds reasonable at first glance, subtle variations in the camera position from shot to shot usually make it difficult to get clean results. Think of the futile attempt of trying to key smoke in front of a brick wall and using a clean plate of the brick wall as your difference input. Part of the wall's structure is always visible in this keying method. Instead, a Difference Keyer is often used to produce a rough matte that is combined with other nodes to produce a more detailed matte.

Inputs

The Difference Keyer node includes four inputs in the Node Editor.

Background: The orange background input accepts a 2D image that contains just the set without your subject.

Foreground: The green foreground input accepts a 2D image that contains the shot with your subject in the frame.

Garbage Matte: The gray garbage matte input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps masks. Connecting a mask to this input causes areas of the image that fall within the matte to be made transparent.

Solid Matte: The white solid matte input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps masks. Connecting a mask to this input causes areas of the image that fall within the matte to be fully opaque.

Effect Mask: The optional blue input expects a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps masks. Connecting a mask to this input limits the pixels where the difference matte occurs. An effects mask is applied to the tool after the tool is processed.

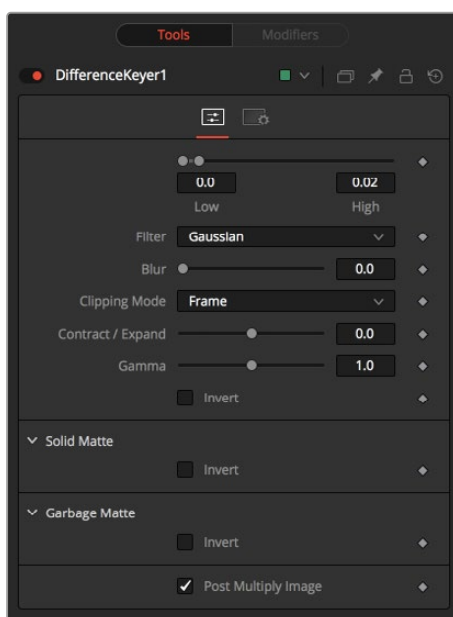
Basic Node Setup

When you do not have content shot on a blue or green screen, the Difference Keyer can be one node in a chain of many used to extract an object from the background. The example below has the Medialn1 as the main subject and a clean background shot without the subject (Background). A B-Spline is used to limit the area the Difference Keyer must deal with for extraction. The result is a matte that can be used to help but not solve the key.



A Difference Keyer with two inputs: one of the subject on a background and the other of just the background

Inspector



The Difference Keyer Controls tab

Controls Tab

The Controls tab in the Difference Keyer contains all the parameters for adjusting the quality of the matte.

Threshold

This range slider sets the lower threshold using the handle on the left and sets the upper threshold using the handle on the right. Adjusting them defines a range of difference values between the images to create a matte.

A difference below the lower threshold setting becomes black or transparent in the matte.

Any difference above the upper threshold setting becomes white or opaque in the matte.

The difference values in the range in between create a grayscale matte.

Filter

This control selects the filtering algorithm used when applying a blur to the matte.

- **Box:** This is the fastest method but at reduced quality. Box is best suited for minimal amounts of blur.
- **Bartlett:** Otherwise known as a Pyramid filter, Bartlett makes a good compromise between speed and quality.
- **Multi-box:** When selecting this filter, the Num Passes slider appears and lets you control the quality. At 1 and 2 passes, results are identical to Box and Bartlett, respectively. At 4 passes and above, results are usually as good as Gaussian, in less time and with no edge “ringing.”
- **Gaussian:** The Gaussian filter uses a true Gaussian approximation and gives excellent results, but it is a little slower than the other filters. In some cases, it can produce an extremely slight edge “ringing” on floating-point pixels.

Blur

This blurs the edge of the matte using the method selected in the Filter menu. A value of zero results in a sharp, cutout-like hard edge. The higher the value, the more blur.

Clipping Mode

This option determines how edges are handled when performing domain of definition rendering. This is profoundly important when blurring the matte, which may require samples from portions of the image outside the current domain.

- **Frame:** The default option is Frame, which automatically sets the node’s domain of definition to use the full frame of the image, effectively ignoring the current domain of definition. If the upstream DoD is smaller than the frame, the remaining area in the frame is treated as black/transparent.
- **Domain:** Setting this option to Domain respects the upstream domain of definition when applying the node’s effect. This can have adverse clipping effects in situations where the node employs a large filter.
- **None:** Setting this option to None does not perform any source image clipping at all. This means that any data required to process the node’s effect that would usually be outside the upstream DoD is treated as black/transparent.

Contract/Expand

This slider shrinks or grows the semitransparent areas of the matte. Values above 0.0 expand the matte, while values below 0.0 contract it.

This control is usually used in conjunction with the blur to take the hard edge of a matte and reduce fringing. Since this control affects only semitransparent areas, it has no effect on a matte’s hard edge.

Gamma

Matte Gamma raises or lowers the values of the matte in the semitransparent areas. Higher values cause the gray areas to be more opaque, and lower values cause the gray areas to be more transparent. Wholly black or white regions of the matte remain unaffected.

Invert

Selecting this checkbox inverts the matte, causing all transparent areas to be opaque and all opaque areas to be transparent.

Solid Matte

Solid Mattes are mask nodes or images connected to the solid matte input on the node. The solid matte is applied directly to the alpha channel of the image. Generally, solid mattes are used to hold out keying in areas you want to remain opaque, such as someone with blue eyes against a blue screen.

Enabling Invert, inverts the solid matte before it is combined with the source alpha.

Garbage Matte

Garbage mattes are mask nodes or images connected to the garbage matte input on the node. The garbage matte is applied directly to the alpha channel of the image. Generally, garbage mattes are used to remove unwanted elements that cannot be keyed, such as microphones and booms. They are also used to fill in areas that contain the color being keyed but that you wish to maintain.

Garbage mattes of different modes cannot be mixed within a single tool. A Matte Control node is often used after a Keyer node to add a garbage matte with the opposite effect of the matte applied to the keyer.

Enabling Invert inverts the garbage matte before it is combined with the source alpha.

Post-Multiply Image

Select this option to cause the keyer to multiply the color channels of the image against the alpha channel it creates for the image. This option is usually enabled and is on by default.

Deselect this checkbox and the image can no longer be considered premultiplied for purposes of merging it with other images. Use the Subtractive option of the Merge node instead of the Additive option.

For more information on these Merge node settings, see *Chapter 95, "Composite Nodes,"* in the DaVinci Resolve Reference Manual, or Chapter 35 in the Fusion Reference Manual.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Matte nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Luma Keyer [LKy]



The Luma Keyer node

Luma Keyer Node Introduction

The Luma Keyer node uses the overall luminance of an image to create an Alpha channel. The label of this node may seem misleading since it allows pulling mattes from almost any channel. In some respects, it is more accurate to call this node an all-purpose channel keyer, but its primary purpose is for extracting alpha channels based on luminance

Inputs

The Luma Keyer node includes four inputs in the Node Editor.

Input: The orange input accepts a 2D image that contains the luminance values you want to be keyed for transparency.

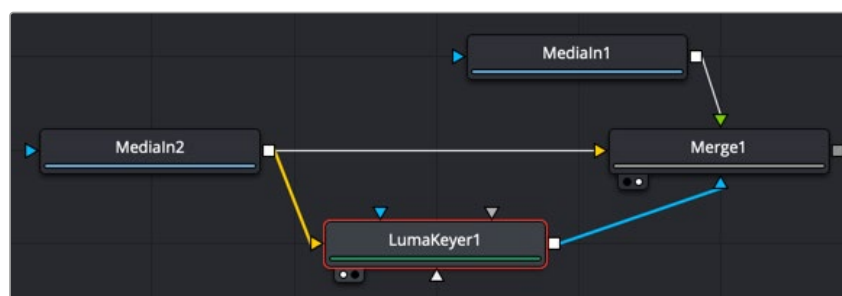
Garbage Matte: The gray garbage matte input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps masks. Connecting a mask to this input causes areas of the image that fall within the matte to be made transparent. The garbage matte is applied directly to the alpha channel of the image.

Solid Matte: The white solid matte input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps masks. Connecting a mask to this input causes areas of the image that fall within the matte to be fully opaque.

Effect Mask: The optional blue input expects a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps masks. Connecting a mask to this input limits the pixels where the luminance key occurs. An effects mask is applied to the tool after the tool is processed.

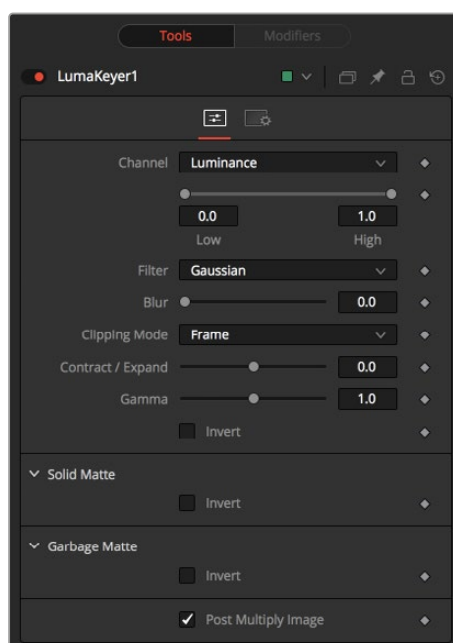
Basic Node Setup

The Luma Keyer takes an input from an image with significant luminance difference to extract a key. You can then use the output of the Luma Keyer into any mask input.



A Luma Keyer output connecting into an effect mask on a Merge node

Inspector



The Luma Keyer Controls tab

Controls Tab

The Controls tab in the Luma Keyer contains all the parameters for adjusting the quality of the matte.

Channel

This menu selects the color channel used for creating the matte. Select from the Red, Green, Blue, Alpha, Hue, Luminance, Saturation, and Depth (Z-buffer) channels.

Threshold

This range slider sets the lower threshold using the handle on the left and sets the upper threshold using the handle on the right. Adjusting them defines a range of luminance values to create a matte.

A value below the lower threshold setting becomes black or transparent in the matte.

Any value above the upper threshold setting becomes white or opaque in the matte.

The values within the range create a grayscale matte.

Filter

This control selects the filtering algorithm used when applying a blur to the matte.

- **Box:** This is the fastest method but at reduced quality. Box is best suited for minimal amounts of blur.
- **Bartlett:** Otherwise known as a Pyramid filter, Bartlett makes a good compromise between speed and quality.
- **Multi-box:** When selecting this filter, the Num Passes slider appears and lets you control the quality. At 1 and 2 passes, results are identical to Box and Bartlett, respectively. At 4 passes and above, results are usually as good as Gaussian, in less time and with no edge “ringing.”
- **Gaussian:** The Gaussian filter uses a true Gaussian approximation and gives excellent results, but it is a little slower than the other filters. In some cases, it can produce an extremely slight edge “ringing” on floating-point pixels.

Blur

This blurs the edge of the matte using the method selected in the Filter menu. A value of zero results in a sharp, cutout-like hard edge. The higher the value, the more blur.

Clipping Mode

This option determines how edges are handled when performing domain of definition rendering. This is profoundly important when blurring the matte, which may require samples from portions of the image outside the current domain.

- **Frame:** The default option is Frame, which automatically sets the node’s domain of definition to use the full frame of the image, effectively ignoring the current domain of definition. If the upstream DoD is smaller than the frame, the remaining area in the frame is treated as black/transparent.
- **Domain:** Setting this option to Domain respects the upstream domain of definition when applying the node’s effect. This can have adverse clipping effects in situations where the node employs a large filter.
- **None:** Setting this option to None does not perform any source image clipping at all. This means that any data required to process the node’s effect that is usually outside the upstream DoD is treated as black/transparent.

Contract/Expand

This slider shrinks or grows the semitransparent areas of the matte. Values above 0.0 expand the matte, while values below 0.0 contract it.

This control is usually used in conjunction with the blur to take the hard edge of a matte and reduce fringing. Since this control affects only semitransparent areas, it has no effect on a matte's hard edge.

Gamma

Matte Gamma raises or lowers the values of the matte in the semitransparent areas. Higher values cause the gray areas to be more opaque, and lower values cause the gray areas to be more transparent. Wholly black or white regions of the matte remain unaffected.

Invert

Selecting this checkbox inverts the matte, causing all transparent areas to be opaque and all opaque areas to be transparent.

Solid Matte

Solid mattes are mask nodes or images connected to the solid matte input on the node. The solid matte is applied directly to the alpha channel of the image. Generally, solid mattes are used to hold out keying in areas you want to remain opaque, such as someone with blue eyes against a blue screen.

Enabling Invert inverts the solid matte before it is combined with the source alpha.

Garbage Matte

Garbage mattes are mask nodes or images connected to the garbage matte input on the node. The garbage matte is applied directly to the alpha channel of the image. Generally, garbage mattes are used to remove unwanted elements that cannot be keyed, such as microphones and booms. They are also used to fill in areas that contain the color being keyed but that you wish to maintain.

Garbage mattes of different modes cannot be mixed within a single tool. A Matte Control node is often used after a Keyer node to add a garbage matte with the opposite effect of the matte applied to the keyer.

Enabling Invert inverts the garbage matte before it is combined with the source alpha.

Post-Multiply Image

Select this option to cause the keyer to multiply the color channels of the image against the alpha channel it creates for the image. This option is usually enabled and is on by default.

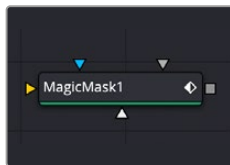
Deselect this checkbox and the image can no longer be considered premultiplied for purposes of merging it with other images. Use the Subtractive option of the Merge node instead of the Additive option.

For more information on these Merge node settings, see *Chapter 95, "Composite Nodes,"* in the DaVinci Resolve Reference Manual, or Chapter 35 in the Fusion Reference Manual.

Common Controls**Settings Tab**

The Settings tab in the Inspector is also duplicated in other Matte nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

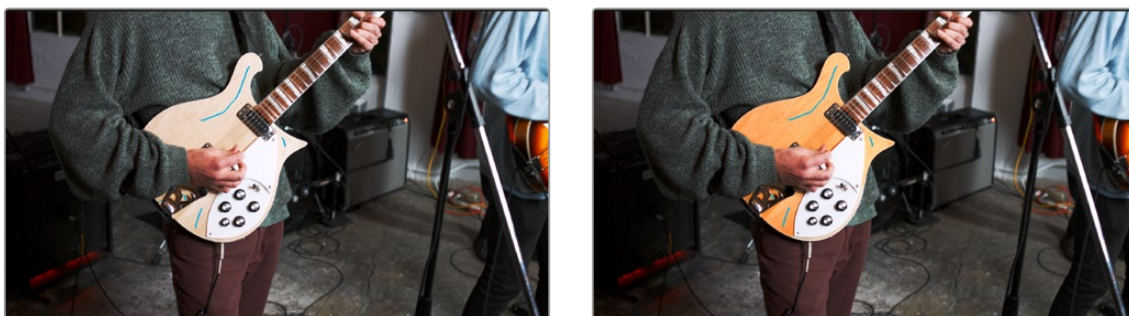
Magic Mask [MagM]



The Magic Mask node

Magic Mask Node Introduction

The Magic Mask palette uses the DaVinci Neural Engine to automatically create a mask to isolate an object in the frame, guided by user-applied paint strokes to identify the subject for isolation. Masks can be generated for either an entire object or just a part of it. This allows you to create complicated masks of complex shapes that may be difficult to isolate with other tools. For example, using the Magic Mask to modify only the wood grain section of this guitar.



(Left) Multiple strokes isolating the wood grain of the guitar, while ignoring the musician's arm; (Right) The finished shot, with a warmer wood grain

NOTE: The Magic Mask tool in Fusion is based on the Object mode of the Magic Mask tool in the Color page and is designed to be used for any subject that is not a person. Subjects like cars, pets, and food are all good candidates for Object mode. If you are trying to isolate a human being, using Magic Mask's Person mode in the Color page will likely give you better results.

Drawing Strokes in the Viewer

The Magic Mask tool works by drawing lines directly into the viewer. Once a stroke is drawn, the viewer will automatically show the mask you've created. You can draw multiple strokes to create a mask in an additive fashion and use negative strokes to remove areas from the mask. It is best to choose a frame to draw on where the object is completely in view and unobstructed.

- **To draw a positive stroke to add an object to a mask:** Load the Magic Mask node in a viewer, and left-click and drag the stroke across the object you want to choose. Alternatively you can change the Stroke Mode to Add in the Inspector and left-click and drag to draw a new positive stroke. Positive strokes are colored blue.

- **To draw a negative stroke to remove an object from a mask:** Load the Magic Mask node in a viewer, and Option-left-click and drag the stroke across the object you want to remove. Alternatively you can change the Stroke Mode to Subtract in the Inspector and left-click and drag to draw a new negative stroke. Negative strokes are colored red.
- **To delete a stroke (or group of strokes) to remove them from the mask:** Shift-drag around a stroke or group of strokes to select them (green), and press the delete key to remove them. Alternatively you can change the Stroke Mode to Select in the Inspector and left-click and drag a selection window to select one or more strokes. Clicking the delete button removes them.

Drawing positive strokes will select areas of similar contrast and color, allowing you to link complex shapes together. Generally you will need more strokes to actively define a complex object, due to the greater variety of the shapes involved. Stroke position is usually more important than stroke length.

Drawing negative strokes removes areas from the object that you don't want to isolate. This can be something simple, like removing the wheels of a car from a mask, or more complicated like removing specific books from a mask of a bookshelf. Stroke position is usually more important than stroke length.



(Left) Multiple strokes isolating the car's body, while removing the wheels and cabin; (Right) The finished shot, fed through a color corrector node to change the color of just the car's body

Inputs

The Magic Mask node includes four inputs in the Node Editor.

Input: The orange input accepts a 2D image that contains the luminance values you want to be keyed for transparency.

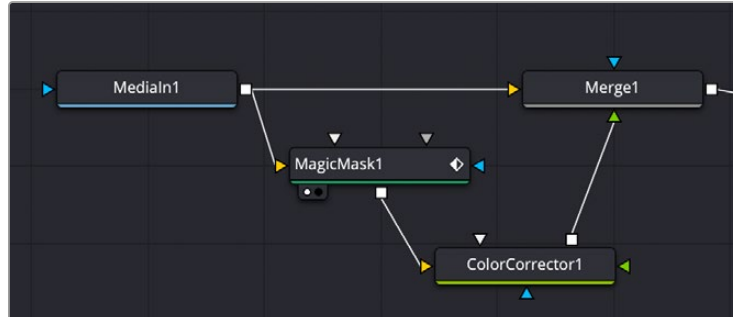
Garbage Matte: The gray garbage matte input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps masks. Connecting a mask to this input causes areas of the image that fall within the matte to be made transparent. The garbage matte is applied directly to the Alpha channel of the image.

Solid Matte: The white solid matte input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps masks. Connecting a mask to this input causes areas of the image that fall within the matte to be fully opaque.

Effect Mask: The optional blue input expects a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps masks. Connecting a mask to this input limits the pixels where the luminance key occurs. An effects mask is applied to the tool after the tool is processed.

Basic Node Setup

The Magic Mask takes RGB inputs from other nodes and lets you isolate specific objects in the frame to use as a mask. Tracking the mask as it changes over time is built into the tool. You can then use the output of the Magic Mask into any mask input.



A Magic Mask node, feeding a matte to a Color Corrector connected to the foreground of a Merge node

Inspector



The Magic Mask Tracking tab

Tracking Tab

You can think of each stroke you've drawn as a persistent eyedropper that samples the image that overlaps it. The mask that results from all strokes' collective analysis of the image is generated live over each frame of the clip. This means that if the camera or the subject moves, you need to motion track or otherwise adjust the position of each stroke to follow along with the motion, so the subject continues to be correctly identified. You also need to make sure that each stroke is able to analyze what it's supposed to, and turn off strokes that can't for whatever reason.

- **Tracking Controls:** These buttons control the tracking direction, from left to right:

Track Reverse: Continuously tracks from the current frame all the way to the beginning of the clip.

Track Reverse One Frame: Tracks one frame backwards and stops. Useful if you're tracking frame-by-frame to watch the progress of a particularly complicated bit of motion. If something goes wrong, you can back up to the last frame where the stroke was able to properly track the subject, and drag the stroke to a better location using the pointer to make it follow the subject properly. If necessary, you can go a frame at a time, dragging the stroke to a better position every time it fails to follow the feature you're using it to isolate.

Stop Tracking: Stops tracking, in cases where there's a problem with the track, and you want to make a change.

Track Forward Then Reverse: Tracks from the current frame all the way to the end of the clip, then returns to the original tracking point and tracks backwards to the beginning of the clip.

Track Forward One Frame: Tracks one frame forward and stops. Useful if you're tracking frame-by-frame to watch the progress of a particularly complicated bit of motion. If necessary, you can go a frame at a time, dragging the stroke to a better position every time it fails to follow the feature you're using it to isolate.

Track Forward: Continuously tracks from the current frame all the way to the end of the clip.

- **Go To Frame:** These buttons snap the playhead to the selected frame, from left to right:

First Frame of Tracked Area: Moves the playhead to the first tracked frame of a range of tracked frames in preparation for tracking backwards if there are untracked frames at the beginning of the clip.

Reference Frame: Moves the playhead to the frame on which you initially drew the strokes.

Last Frame of Tracked Area: Moves the playhead to the last tracked frame of a range of tracked frames in preparation for tracking forwards if there are untracked frames at the end of the clip.

- **Stroke Mode:** These buttons let you change and modify the strokes drawn in the Viewer

Add: Lets you add an additional stroke (blue) that determines what in the frame will be included in the mask.

Subtract: Lets you add an additional stroke (red) that determines what in the frame will be excluded from the mask.

Select: Lets you draw a selection rectangle around single or multiple strokes to select them (green).

Delete: Deletes all selected strokes (green).

- **Clear Strokes:** These buttons remove all strokes from the selected range.

Current: Removes strokes from the frame under the playhead.

Range: Removes strokes from the selected render range.

All: Removes all strokes from the entire clip.

- **Disk Cache:** These buttons allow you to control what's stored in the disk cache.

Regenerate All: Rebuilds the disk cache.

Clear: Deletes the frames in the current cache.

- **Reference Time:** The Reference Time determines the frame where the initial strokes are drawn. It is also the time from which tracking begins. The reference frame cannot be changed once it has been set without destroying all pre-existing tracking information.
- **Processed Frames:** Displays the range of frames that have already been tracked. Start is the earliest frame tracked and end is the last frame tracked. These fields are not user editable.
- **Mode:** Two options let you choose a tradeoff between quality and performance. Faster lets you generate a lower quality mask more quickly that's suitable for garbage matting. Better generates a higher quality mask with more detail that's more processor-intensive.

Post-Multiply Image

Select this option to cause the keyer to multiply the color channels of the image against the Alpha channel it creates for the image. This option is usually enabled and is on by default.

Deselect this checkbox and the image can no longer be considered premultiplied for purposes of merging it with other images. Use the Subtractive option of the Merge node instead of the Additive option.

For more information on these Merge node settings, see *Chapter 95, "Composite Nodes,"* in the DaVinci Resolve Reference Manual, or Chapter 35 in the Fusion Reference Manual.

Matte Tab

The Matte tab refines the alpha of the key, combined with any solid and garbage masks connected to the node. When using the Matte tab, set the viewer to display the Alpha channel of Magic Mask's final output.

- **Filter:** Selects the Filter that is used when blurring the matte.
 - Box Blur:** This option applies a Box Blur effect to the whole image. This method is faster than the Gaussian blur but produces a lower-quality result.
 - Bartlett:** Bartlett applies a more subtle, anti-aliased blur filter.
 - Multi-Box:** Multi-Box uses a box filter layered in multiple passes to approximate a Gaussian shape. With a moderate number of passes (e.g., four), a high-quality blur can be obtained, often faster than the Gaussian filter and without any ringing.
 - Gaussian:** Gaussian applies a smooth, symmetrical blur filter, using a sophisticated constant-time Gaussian approximation algorithm.
 - Fast Gaussian:** Gaussian applies a smooth, symmetrical blur filter, using a sophisticated constant-time Gaussian approximation algorithm. This mode is the default filter method.
- **Blur:** This blurs the edge of the matte using a standard, constant speed Gaussian blur. A value of zero results in a sharp, cutout-like hard edge. The higher the value, the more blur is applied to the matte.
- **Erode/Dilate:** This shrinks or grows the matte. Eroding the matte reveals more of the foreground input, while dilating the matte reveals more of the background input. Values above 0.0 dilate the matte and values below 0.0 erode it.
- **Gamma:** This raises or lowers the values of the matte in the semitransparent areas. Higher values cause the gray areas to become more opaque, and lower values cause the gray areas to become more transparent. Completely black or white regions of the matte remain unaffected.
- **Threshold:** Any value below the lower threshold becomes black or transparent in the matte. Any value above the upper threshold becomes white or opaque in the matte. All values within the range maintain their relative transparency values.
- **Restore Fringe:** This restores the edge of the matte around the keyed subject. Often when keying, the edge of the subject where you have hair is clipped out. Restore Fringe brings back that edge while keeping the matte solid.
- **Invert Matte:** When this checkbox is selected, the Alpha channel of the image is inverted, causing all transparent areas to be opaque and all opaque areas to be transparent.

Solid Matte

Solid mattes are mask nodes or images connected to the solid matte input on the node. The solid matte is applied directly to the Alpha channel of the image. Generally, solid mattes are used to hold out keying in areas you want to remain opaque, such as someone with blue eyes against a blue screen.

Enabling Invert inverts the solid matte before it is combined with the source alpha.

Garbage Matte

Garbage mattes are mask nodes or images connected to the garbage matte input on the node. The garbage matte is applied directly to the Alpha channel of the image. Generally, garbage mattes are used to remove unwanted elements that cannot be keyed, such as microphones and booms. They are also used to fill in areas that contain the color being keyed but that you wish to maintain.

Garbage mattes of different modes cannot be mixed within a single tool. A Matte Control node is often used after a Keyer node to add a garbage matte with the opposite effect of the matte applied to the keyer.

Enabling Invert inverts the garbage matte before it is combined with the source alpha.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Matte nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Matte Control [Mat]



The Matte Control node

Matte Control Node Introduction

Keyer nodes are used to create an alpha channel on an image that does not have one. The Matte Control node is used to combine and manipulate the alpha channels embedded in images as well as masks created by masking tools.

Typically, you add this node to copy a color channel or alpha channel from the foreground input to the background input, or to combine alpha channels from the two images.

Inputs

The Matte Control node includes four inputs in the Node Editor.

Background: The orange background input accepts a 2D image that receives the foreground image alpha channel (or some other channel you want to copy to the background).

Foreground: The green foreground input accepts a 2D image that contains an alpha channel (or some other channel) you want to be applied to the background image.

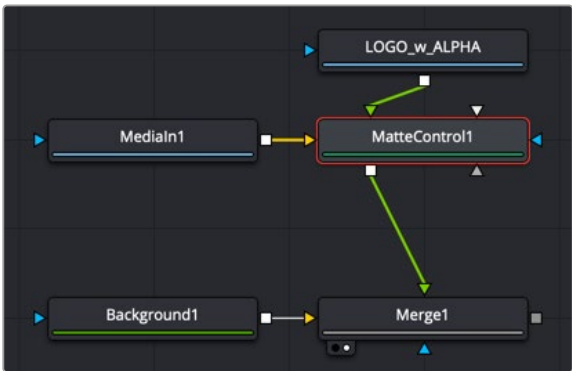
Garbage Matte: The gray garbage matte input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps masks. Connecting a mask to this input causes areas of the foreground/background combination that fall within the matte to be made transparent.

Solid Matte: The white solid matte input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps masks. Connecting a mask to this input causes areas of the foreground/background combination that fall within the matte to be fully opaque.

Effect Mask: The optional blue input expects a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps masks. Connecting a mask to this input limits the pixels where the matte control occurs. An effects mask is applied to the tool after the tool is processed.

Basic Node Setup

Below, the Matte Control node is set up to copy the foreground (green) input's alpha channel into the background (orange) input. The output of the Matte Control is then an image with an alpha channel used as the foreground composite in the Merge node.



A Matte Control embedded an alpha from the foreground input to the background input

Inspector



The Matte Control Matte tab

Matte Tab

The Matte tab combines and modifies alpha or color channels from an image in the foreground input with the background image.

Combine

Use this menu to select which operation is applied. The default is set to None for no operation.

- **None:** This causes the foreground image to be ignored.
- **Combine Red:** This combines the foreground red channel to the background alpha channel.
- **Combine Green:** This combines the foreground green channel to the background alpha channel.
- **Combine Blue:** This combines the foreground blue channel with the background alpha channel.
- **Combine Alpha:** This combines the foreground alpha channel with the background alpha channel.
- **Solid:** This causes the background alpha channel to become completely opaque.
- **Clear:** This causes the background alpha channel to become completely transparent.

Combine Operation

Use this menu to select the method used to combine the foreground channel with the background.

- **Copy:** This copies the foreground source over the background alpha, overwriting any existing alpha in the background.
- **Add:** This adds the foreground source to the background alpha.
- **Subtract:** This subtracts the foreground source from the background alpha.
- **Inverse Subtract:** This subtracts the background alpha from the foreground source.
- **Maximum:** This compares the foreground source and the background alpha and takes the value from the pixel with the highest value.
- **Minimum:** This compares the foreground source and the background alpha and takes the value from the pixel with the lowest value.
- **And:** This performs a logical AND on the two values.
- **Or:** This performs a logical OR on the values.
- **Merge Over:** This merges the foreground source channel over the background alpha channel.
- **Merge Under:** This merges the foreground source channel under the background alpha channel.

Filter

Selects the Filter that is used when blurring the matte.

- **Box Blur:** This option applies a Box Blur effect to the whole image. This method is faster than the Gaussian blur but produces a lower-quality result.
- **Bartlett:** Bartlett applies a more subtle, anti-aliased blur filter.
- **Multi-Box:** Multi-Box uses a box filter layered in multiple passes to approximate a Gaussian shape. With a moderate number of passes (e.g., four), a high-quality blur can be obtained, often faster than the Gaussian filter and without any ringing.
- **Gaussian:** Gaussian applies a smooth, symmetrical blur filter, using a sophisticated constant-time Gaussian approximation algorithm. In extreme cases, this algorithm may exhibit ringing; see below for a discussion of this. This mode is the default filter method.

Blur

This blurs the edge of the matte using a standard, constant speed Gaussian blur. A value of zero results in a sharp, cutout-like hard edge. The higher the value, the more blur is applied to the matte.

Clipping Mode

This option determines how edges are handled when performing domain of definition rendering. This is profoundly important when blurring the matte, which may require samples from portions of the image outside the current domain.

- **Frame:** The default option is Frame, which automatically sets the node's domain of definition to use the full frame of the image, effectively ignoring the current domain of definition. If the upstream DoD is smaller than the frame, the remaining area in the frame is treated as black/transparent.
- **Domain:** Setting this option to Domain respects the upstream domain of definition when applying the node's effect. This can have adverse clipping effects in situations where the node employs a large filter.
- **None:** Setting this option to None does not perform any source image clipping at all. This means that any data required to process the node's effect that is usually outside the upstream DoD is treated as black/transparent.

Contract/Expand

This shrinks or grows the matte similar to an Erode Dilate node. Contracting the matte reveals more of the foreground input, while expanding the matte reveals more of the background input. Values above 0.0 expand the matte, and values below 0.0 contract it.

Gamma

This raises or lowers the values of the matte in the semitransparent areas. Higher values cause the gray areas to become more opaque, and lower values cause the gray areas to become more transparent. Completely black or white regions of the matte remain unaffected.

Threshold

Any value below the lower threshold becomes black or transparent in the matte. Any value above the upper threshold becomes white or opaque in the matte. All values within the range maintain their relative transparency values.

Restore Fringe

This restores the edge of the matte around the keyed subject. Often when keying, the edge of the subject where you have hair is clipped out. Restore Fringe brings back that edge while keeping the matte solid.

Invert Matte

When this checkbox is selected, the alpha channel of the image is inverted, causing all transparent areas to be opaque and all opaque areas to be transparent.

Solid Matte

Solid mattes are mask nodes or images connected to the solid matte input on the node. The solid matte is applied directly to the alpha channel of the image. Generally, solid mattes are used to hold out areas you want to remain opaque, such as someone with blue eyes against a blue screen.

Enabling Invert inverts the solid matte before it is combined with the source alpha.

Garbage Matte

Garbage mattes are mask nodes or images connected to the garbage matte input on the node. The garbage matte is applied directly to the alpha channel of the image. Generally, garbage mattes are used to remove unwanted elements that cannot be keyed, such as microphones and booms. They are also used to fill in areas that contain the color being keyed but that you wish to maintain.

Garbage mattes of different modes cannot be mixed within a single tool. A Matte Control node is often used after a Keyer node to add a garbage matte with the opposite effect of the matte applied to the keyer.

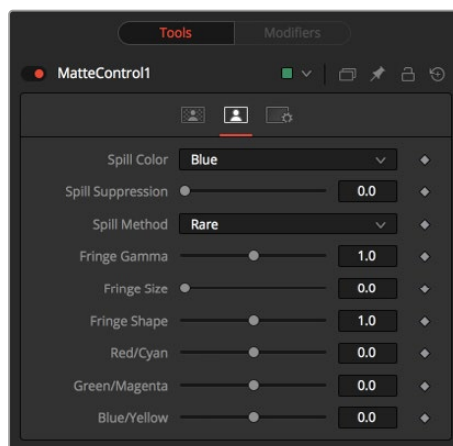
Enabling Invert inverts the garbage matte before it is combined with the source alpha.

Post-Multiply Image

Selecting this option multiplies the color channels of the image against the alpha channel it creates for the image. This option is usually enabled and is on by default.

Deselect this checkbox and the image can no longer be considered premultiplied for purposes of merging it with other images. Use the Subtractive option of the Merge node instead of the Additive option.

For more information on these Merge node settings, see *Chapter 95, "Composite Nodes,"* in the DaVinci Resolve Reference Manual, or Chapter 35 in the Fusion Reference Manual.



The Matte Control Spill tab

Spill Tab

The Spill tab handles spill suppression in the Matte Control. Spill suppression is a form of color correction that attempts to remove the screen color from the fringe of the matte.

Spill is the transmission of the screen color through the semitransparent areas of the alpha channel. In the case of blue- or green-screen keying, this usually causes the color of the background to become apparent in the edges of the foreground subject.

Spill Color

This menu selects the color used as the base for all spill suppression techniques.

Spill Suppression

When this slider is set to 0, no spill suppression is applied to the image. Increasing the slider increases the strength of the spill method.

Spill Method

This selects the strength of the algorithm used to apply spill suppression to the image.

- **None:** None is selected when no spill suppression is required.
- **Rare:** This removes very little of the spill color and is the lightest of all methods.
- **Medium:** This works best for green screens.
- **Well Done:** This works best for blue screens.
- **Burnt:** This works best for blue screen. Use this mode only for very troublesome shots.

Fringe Gamma

This control can be used to adjust the brightness of the fringe or halo that surrounds the keyed image.

Fringe Size

This expands and contracts the size of the fringe or halo surrounding the keyed image.

Fringe Shape

Fringe Shape presses the fringe toward the external edge of the image or pulls it toward the inner edge of the fringe. Its effect is most noticeable while the Fringe Size value is large.

Cyan/Red, Magenta/Green, and Yellow/Blue

Use these three controls to color correct the fringe of the image.

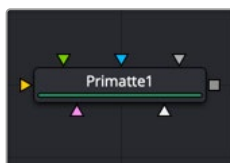
This is useful for correcting semitransparent pixels that still contain color from the original background to match the new background.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Matte nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Primatte [Pri]



The Primatte node

NOTE: Primatte is only available in Fusion Studio.

Primatte Node Introduction

Primatte is an advanced keying tool for Fusion Studio. To use Primatte effectively, you must understand how it works. Using a series of selection buttons, Primatte assigns RGB pixels into one of the four specific zones.

- **Zone 1:** Complete background image.
- **Zone 2:** Foreground image with spill suppression and transparency.
- **Zone 3:** Foreground image with spill suppression only.
- **Zone 4:** Complete foreground image.

Depending on the type of blue- or green-screen content, you may find that the Delta Keyer or the Primatte keyer handles the specific keying task better. There is no one-solution-fits-all when it comes to keying, and in some cases, the combination of the two keyers may prove to be the best solution.

NOTE: Primatte is distributed and licensed by IMAGICA Corp. of America, Los Angeles, CA, USA. Primatte was developed by and is a trademark of IMAGICA Corp., Tokyo, Japan.

Inputs

The Primatte node includes six inputs in the Node Editor. Unlike every other tool in Fusion, the primary orange input is labeled as the Foreground input, since it accepts the green-screen or blue-screen image. The background input on the Primatte node is the green input; this is an optional input that allows Primatte to create the final merged composite.

Foreground Input: The orange input accepts a 2D image that contains blue or green screen.

Background Input: The green (optional) input accepts a 2D image layered as the background in the composite. If no image is connected, Primatte outputs the keyed foreground. Connecting an image to the background input activates Primatte's advanced edge blending options.

Replacement Image: The magenta (optional) input accepts a 2D image used as a source of Primatte's spill suppression color correction.

Garbage Matte: The gray garbage matte input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps masks. Connecting a mask to this input causes areas of the image that fall within the matte to be made transparent. The garbage matte is applied directly to the alpha channel of the image.

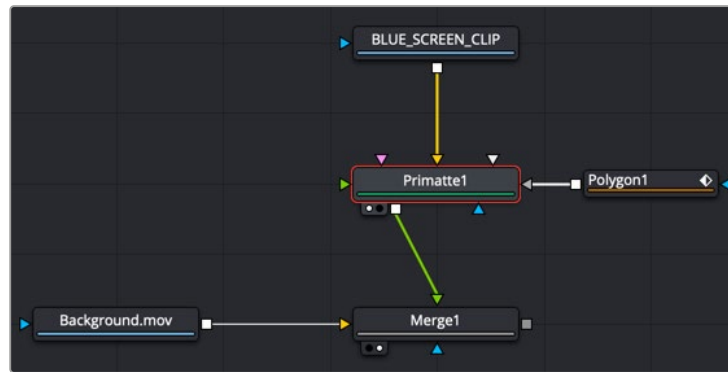
Solid Matte: The white solid matte input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps masks. Connecting a mask to this input causes areas of the image that fall within the matte to be fully opaque.

Effect Mask: The optional blue input expects a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps masks. Connecting a mask to this input limits the pixels where the keying occurs. An effects mask is applied to the tool after the tool is processed.

NOTE: Connecting the background input without connecting the replacement image input uses the background image as the replacement image surf spill suppression.

Basic Node Setup

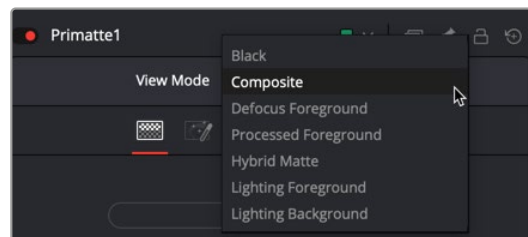
A single Primatte keyer can rarely get perfect results because most green- or blue-screen shots have problems the keyer is not made to handle. Keyers often need the help of garbage mattes or solid mattes created with a Polygon or B-Spline node. Shots can also require more than just one keyer to achieve perfect results. Below, the Primatte node has the blue-screen content connected to the orange input. Unlike other Fusion nodes, the foreground gets connected to the orange input. The result is an image with alpha that can then be connected into the foreground of a Merge node.



A Primatte node combined with a polygon matte as a garbage matte and connected to the foreground of a Merge

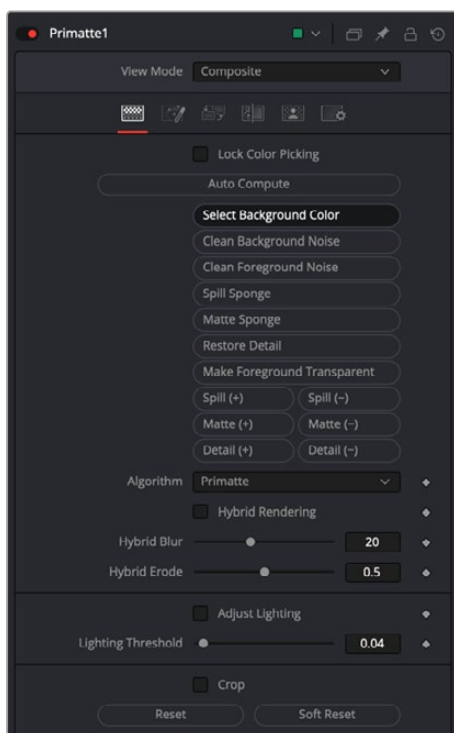
Primatte Tab View Mode

At the top of the Inspector is the View Mode menu. The default selection shows the final Composite result. You can change the view to see various intermediate stages of the keying process.



The Primatte tab View mode

- **Black:** Displays the foreground subject on a black or transparent background.
- **Composite:** The final keyed image with spill suppression, composited over the image connected to the green Background Input on the node.
- **Defocus Foreground:** Displays the output of the Pre Matte key.
- **Processed Foreground:** Displays the alpha of the key before being combined with solid and garbage masks. When displaying the matte, set the viewer to show the alpha channel.
- **Hybrid Matte:** Displays the matte generated when the Hybrid Rendering checkbox is enabled. Best viewed when adjusting the Hybrid Blur and Hybrid Erode sliders.
- **Lighting Foreground:** Displays the foreground subject over the optimized artificial backing screen that the Adjust Lighting mode creates.
- **Lighting Background:** Displays the optimized artificial backing screen that the Adjust Lighting mode creates.



The Primatte tab Inspector

Primatte Tab

The core functionality for Primatte is found in the Primatte tab. The basic workflow is based on selecting one of the operational mode buttons and then scrubbing over areas in the viewer.

Lock Color Picking

Activate this button once you finished adjusting your key to prevent making accidental changes in the viewer.

Auto Compute

The Auto Compute button is likely the first button pressed when starting to key your footage. Primatte automatically analyzes the original foreground image, determines the backing color, and sets it as the central backing color. Then, using that information, another analysis determines the foreground areas. A Clean FG Noise operation is performed using the newly determined foreground areas, and Primatte renders the composite.

NOTE: The Auto Compute button may make the next three buttons—Select Background Color, Clean Background Noise, and Clean Foreground Noise—unnecessary and make your keying operation much more straightforward. Clicking Auto Compute automatically senses the backing screen color, eliminates it, and even gets rid of some foreground and background noise. If you get good results, then jump ahead to the Spill Removal tools. If you don't get satisfactory results, continue from this point using the three buttons described below.

Select Background Color

Clicking the Select Background Color button allows you to select the screen color by scrubbing in the viewer. It uses the traditional Primatte method of taking the sampled backing screen color, projecting

a line in the opposite direction on the hue wheel, and generating artificial pixels that may represent the FG object. Then, using the artificially generated foreground pixels, it internally does the Clean FG Noise operation and creates the shape of the middle and outer polyhedrons. It then renders the composite using these generated polyhedrons. This does not automatically use the Adjust Lighting functionality, as it must be selected in a separate operation.

Clean Background Noise

Clicking this button helps to remove any white regions in the dark screen area (“noise”), or shades of the screen color that did not get picked up on the first sample. Once you click the button, scrub the mouse pointer over areas in the viewer to sample white-ish noise regions.

Clean Foreground Noise

If there are dark transparent regions in the middle of the mostly white opaque foreground object, click the Clean Foreground Noise button and scrub over the dark pixels in the foreground area until that area is as white as possible.

Spill Sponge

The Spill Sponge is the quickest method for removing color spill on your subject. Click the Spill Sponge button and scrub the mouse pointer over a screen color pixel, and the screen color disappears from the selected color region and is replaced by a complementary color, a selected color, or a color from a replacement image. These options are set in the Replace tab. Additionally, use the tools under the Fine Tuning tab or use the Spill(+) and Split(-) features to adjust the spill.

Matte Sponge

Sometimes in the Primatte operation, a 100% opaque, foreground area (all white) becomes slightly transparent (gray). To clean those transparent areas, click the Matte Sponge button and scrub over the transparent pixels. All the spill-suppression information remains intact.

Restore Detail

Clicking Restore Detail and scrubbing over background regions in the viewer turns completely transparent areas translucent. This operation is useful for restoring lost hair details, thin wisps of smoke, and the like.

Make Foreground Transparent

When this button is selected, the opaque foreground color region sampled in the viewer becomes slightly translucent. This operation is useful for the subtle tuning of foreground subjects, which are otherwise 100 percent covered with smoke or clouds. It can be used only one time on a particular color. For a more flexible way to thin out a color region, and to be able to take multiple samples, use the Matte(-) tool.

Spill(+)

Clicking the Spill(+) button returns the color spill to the sampled pixel color (and all colors like it) in the amount of one Primatte increment. This tool can be used to move the sampled color more in the direction of the color in the original foreground image. It can be used to nullify a Spill(-) step.

Spill(-)

Clicking the Spill(-) button removes from the sampled pixel color (and all colors like it) in the amount of one Primatte increment. If spill color remains, another click using this operational mode tool removes more of the color spill. Continue using this tool until all color spill has been removed from the sampled color region.

Matte(+)

Clicking the Matte(+) button makes the matte more opaque for the sampled pixel color (and all colors like it) in the amount of one Primatte increment. If the matte is still too translucent or thin, another click using this operational mode tool makes the sampled color region even more opaque. This can be used to thicken smoke or make a shadow darker to match shadows in the background imagery. It can only make these adjustments to the density of the color region on the original foreground image. It can be used to nullify a Matte(-) step.

Matte(-)

Clicking the Matte(+) button makes the matte more translucent for the sampled pixel color (and all colors like it) in the amount of one Primatte increment. If the matte is still too opaque, another click using this operational mode tool makes the sampled color region even more translucent. This can be used to thin out smoke or make a shadow thinner to match shadows in the background imagery.

Detail(+)

When this button is selected, the foreground detail becomes less visible for the sampled pixel color (and all colors like it) in the amount of one Primatte increment. If there is still too much detail, another click using this operational mode tool makes more of it disappear. This can be used to remove smoke or wisps of hair from the composite. Sample where detail is visible, and it disappears. This is for moving color regions into the 100% background region. It can be used to nullify a Detail(-) step.

Detail(-)

When this button is selected, foreground detail becomes more visible for the sampled pixel color (and all colors like it) in the amount of one Primatte increment. If detail is still missing, another click using this operational mode tool makes detail more visible. This can be used to restore lost smoke or wisps of hair. Sample where the smoke or hair just disappears and it returns to visibility. Use this for restoring color regions that were moved into the 100% background region. It may start to bring in background noise if shooting conditions were not ideal on the foreground image.

Algorithms

There are three keying algorithms available in the Primatte keyer:

- **Primatte:** The Primatte algorithm mode delivers the best results and supports both the Solid Color and the Complement Color spill suppression methods. This algorithm uses three multifaceted polyhedrons (as described later in this section) to separate the 3D RGB colorspace. It is also the default algorithm mode and, because it is computationally intensive, it may take the longest to render.
- **Primatte RT:** Primatte RT is the simplest algorithm and therefore the fastest. It uses only a single planar surface to separate the 3D RGB colorspace (as described later in this section) and, as a result, does not separate the foreground from the backing screen as carefully as the above Primatte algorithm. Another disadvantage of the Primatte RT algorithm is that it does not work well with less saturated backing screen colors, and it does not support the Complement Color spill suppression method.
- **Primatte RT+:** Primatte RT+ is in between the above two options. It uses a six planar surface color separation algorithm (as described later in this section) and delivers results in between the other two options in both quality and performance. Another disadvantage of the Primatte RT+ algorithm is that it does not work well with less saturated backing screen colors, and it does not support the Complement Color spill suppression method.

Hybrid Rendering

After sampling the backing screen color and producing acceptable edges around the foreground object, you sometimes find a transparent area within the foreground subject. This can occur when the foreground subject contains a color that is close to the backing screen color. Removing this transparency with the Clean FG Noise mode can cause the edge of the foreground subject to pick up a fringe that is close to the backing screen color. Removing the fringe is very difficult without sacrificing quality somewhere else on the image. The Hybrid Render mode internally creates two keying operations: Body and Edge. The optimized Edge operation gets the best edge around the foreground subject without any fringe effect. The Body operation deals with transparency within the foreground subject. The resultant matte is created by combining these two mattes, and then blurring and eroding the foreground subject in the Body matte and combining it with the edge matte.

To use Hybrid Rendering, start by keying the main foreground area using the Select Background Color mode (or any of the other Primatte backing screen detection methods). Activate the Hybrid Rendering checkbox. Lastly, select the Clean FG Noise button and scrub over the transparent area. The Hybrid Render mode performs the “Body/Edge” operation. The result is a final composite with perfect edges around the foreground subject with a solid foreground subject.

Hybrid Blur

Blurs the Body matte that has been automatically generated when Hybrid Rendering is activated.

Hybrid Erode

This slider dilates or erodes the Hybrid matte. You can view the results by selecting Hybrid matte in the View Mode menu.

Adjust Lighting

Before applying the Adjust Lighting operation, it is necessary to determine the backing screen color using Auto Compute or Select Background Color. After performing one of those operations, click on the Adjust Lighting button. Primatte generates an artificial clean plate and uses it to generate an evenly lit backing screen behind the foreground object. The default setting should detect all the areas that contain foreground pixels and deliver a smooth backing screen for the keying.

Lighting Threshold

Should Adjust Lighting fail to produce a smoother backing screen, adjust the Lighting Threshold slider while viewing the Lighting Background setting in the View Mode menu. This displays the optimized artificial backing screen that the Adjust Lighting mode creates.

Crop

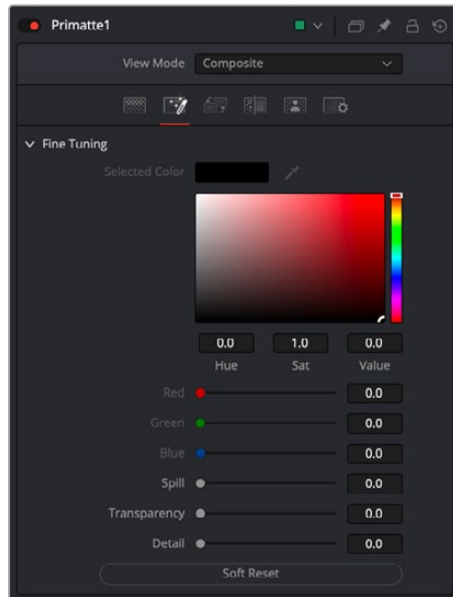
This button reveals the Crop sliders to create a rectangular garbage matte with the Primatte node. As opposed to Fusion’s Crop tool, this does not change the actual image size.

Reset

Resets all the Primatte key control data back to a blue- or green-screen.

Soft Reset

Resets just the Primatte parameters used since the Select Background Color operation was last completed.



The Primatte Fine Tuning tab

Fine Tuning Tab

The Fine Tuning tab can make refined adjustments to the spill suppression, density of the matte, and semitransparent areas. These sliders provide a bit more granularity over the Spill(+)(-), Matte(+)(-) and Detail(+)(-) buttons in the Primatte tab.

Selected Color

This shows the color selected (or registered) by the scrubbing in the viewer while the Fine Tuning tab is selected.

Fine Tuning Sliders

The color of the scrubbed pixel is registered as a reference color for fine tuning. It is displayed in the Color swatch. To perform the tuning operation, sample a color region on the image, and adjust one of the Fine Tuning sliders to achieve the desired effect.

Spill

The Spill slider can be used to remove spill from the selected color region. The more to the right the slider moves, the more spill is removed. The more to the left the slider moves, the closer the color component of the selected region is to the color in the original foreground image. If moving the slider to the right does not remove the spill, resample the color region and move the slider again.

These slider operations are additive. The result achieved by moving the slider to the right can also be achieved by clicking on the color region using the Spill(-) operational mode.

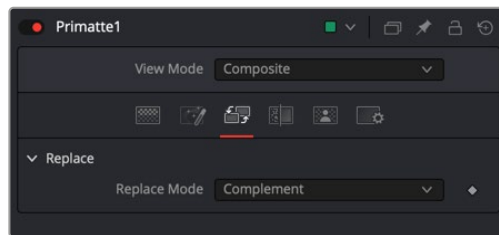
Transparency

The Transparency slider makes the matte more translucent in the selected color region. Moving this slider to the right makes the selected color region more transparent. Moving the slider to the left makes the matte more opaque. If moving the slider to the right does not make the color region translucent enough, resample the color region and again move the slider to the right. These slider operations are additive. The result achieved by moving the slider to the right can also be achieved by clicking on the color region using the Matte(-) operational mode.

Detail

The Detail slider can be used to restore lost detail. After selecting a color region, moving this slider to the left makes the selected color region more visible. Moving the slider to the right makes the color region less visible. If moving the slider to the left does not make the color region visible enough, resample the color region and again move the slider to the left.

These slider operations are additive. This result achieved by moving the slider to the left can also be achieved by clicking on the color region using the Detail(-) operational mode.



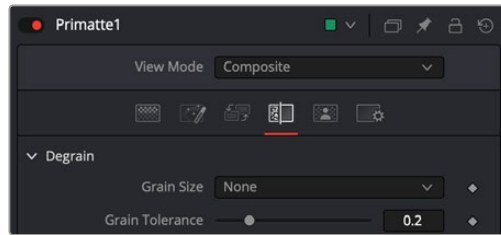
The Primatte Replace tab

Replace Tab

The Replace tab allows you to choose between the three methods of color spill replacement as covered in detail in the Spill Sponge section above. There are three options for the replacement color when removing the spill. These options are selected from the Replace mode menu.

Replace Mode

- **Complement:** Replaces the spill color with the complement of the screen color. This mode maintains fine foreground detail and delivers the best-quality results. If foreground spill is not a significant problem, this mode is the one that should be used. However, If the spill intensity on the foreground image is rather significant, this mode may often introduce serious noise in the resultant composite.
- **Image:** Replaces the spill color with colors from a defocused version of the background image or the Replace image, if one is connected to the Replace input (magenta) on the node. This mode results in a good color tone on the foreground subject even with a high-contrast background. On the negative side, the Image mode occasionally loses the fine edge detail of the foreground subjects. Another problem can occur if you later change the size of the foreground image against the background. Since the background/foreground alignment would change, the applied color tone from the defocused image might not match the new alignment.
- **Color:** Replaces the spill color with a solid color. When this option is selected, a color swatch and R,G,B sliders are displayed for selecting the color. Changing the palette color for the solid replacement, you can select a good spill replacement that matches the composite background. Its strength is that it works fine with even severe spill conditions. On the negative side, when using the Solid Color Replacement mode, fine detail on the foreground edge tends to be lost. The single palette color sometimes cannot make a good color tone if the background image has some high-contrast color areas.



The Primatte Degrain tab

Degrain Tab

The Degrain tab is used when a foreground image is highly compromised by film grain. As a result of the grain, when backing screen noise is completely removed, the edges of the foreground object often become harsh and jagged, leading to a poor key.

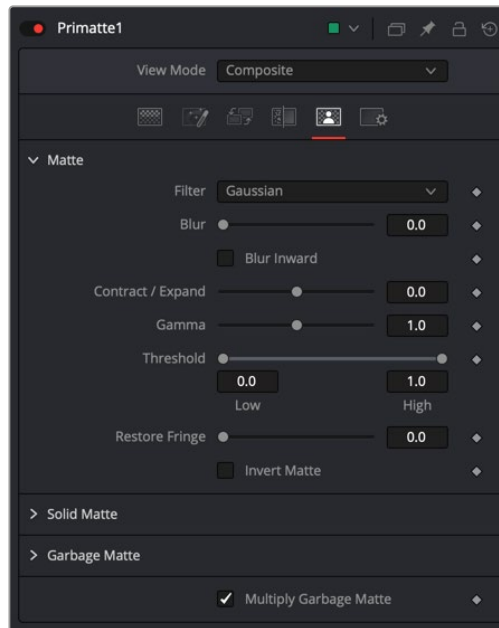
Grain Size

The Grain Size selector provides a range of grain removal from Small to Large. If the foreground image has a large amount of film grain-induced pixel noise, you may lose a good edge to the foreground object when trying to clean all the grain noise with the Clean Background Noise Operation Mode. These tools clean up the grain noise without affecting the quality of the key.

- **None:** No degrading is performed.
- **Small:** The average color of a small region of the area around the sampled pixel. This should be used when the grain is very dense.
- **Medium:** The average color of a medium-sized region of the area around the sampled pixel. This should be used when the grain is less dense.
- **Large:** The average color of a larger region of the area around the sampled pixel. This should be used when the grain is very loose.

Grain Tolerance

Adjusting this slider increases the effect of the Clean Background Noise tool without changing the edge of the foreground object.



The Primatte Matte tab

Matte Tab

The Matte tab refines the alpha of the key, combined with any solid and garbage masks connected to the node. When using the Matte tab, set the viewer to display the alpha channel of Primatte's final output.

Filter

This control selects the filtering algorithm used when applying blur to the matte.

- **Box:** This is the fastest method but at reduced quality. Box is best suited for minimal amounts of blur.
- **Bartlett:** Otherwise known as a Pyramid filter, Bartlett makes a good compromise between speed and quality.
- **Multi-Box:** When selecting this filter, the Num Passes slider appears and lets you control the quality. At 1 and 2 passes, results are identical to Box and Bartlett, respectively. At 4 passes and above, results are usually as good as Gaussian, in less time and with no edge “ringing.”
- **Gaussian:** The Gaussian filter uses a true Gaussian approximation and gives excellent results, but it is a little slower than the other filters. In some cases, it can produce an extremely slight edge “ringing” on floating-point pixels.

Blur

Matte Blur blurs the edge of the matte based on the Filter menu setting. A value of zero results in a sharp, cutout-like hard edge. The higher the value, the more blur applied to the matte.

Blur Inward

Activating the Blur Inward checkbox generates the blur toward the center of the foreground subject. Conventional blurring or defocus affects the matte edges in both directions (inward and outward) and sometimes introduces a halo artifact around the edge in the composite view. Blur Inward functions only in the inward direction of the foreground subject (toward the center of the white area). The final result removes small and dark noise in the screen area without picking them up again in the Clean Background Noise mode. It can sometimes result in softer, cleaner edges on the foreground objects.

Contract/Expand

This slider shrinks or grows the semitransparent areas of the matte. Values above 0.0 expand the matte, while values below 0.0 contract it.

This control is usually used in conjunction with the Matte Blur to take the hard edge of a matte and reduce fringing. Since this control affects only semitransparent areas, it will have no effect on a matte's hard edge.

Gamma

Matte Gamma raises or lowers the values of the matte in the semitransparent areas. Higher values cause the gray areas to become more opaque, and lower values cause the gray areas to become more transparent. Completely black or white regions of the matte remain unaffected.

Since this control affects only semitransparent areas, it will have no effect on a matte's hard edge.

Threshold

This range slider sets the lower threshold using the handle on the left and sets the upper threshold using the handle on the right.

Any value below the lower threshold setting becomes black or transparent in the matte.

Any value above the upper threshold setting becomes white or opaque in the matte. All values within the range maintain their relative transparency values.

This control is often used to reject salt and pepper noise in the matte.

Restore Fringe

This restores the edge of the matte around the keyed subject. Often when keying, the edge of the subject where you have hair is clipped out. Restore Fringe brings back that edge while keeping the matte solid.

Invert Matte

When this checkbox is selected, the alpha channel created by the keyer is inverted, causing all transparent areas to be opaque and all opaque areas to be transparent.

Solid Matte

Solid mattes are mask nodes or images connected to the solid matte input on the node. The solid matte is applied directly to the alpha channel of the image. Generally, solid mattes are used to hold out keying in areas you want to remain opaque, such as someone with blue eyes against a blue screen.

Enabling Invert inverts the solid matte before it is combined with the source alpha.

Garbage Matte

Garbage mattes are mask nodes or images connected to the garbage matte input on the node. The garbage matte is applied directly to the alpha channel of the image. Generally, garbage mattes are used to remove unwanted elements that cannot be keyed, such as microphones and booms. They are also used to fill in areas that contain the color being keyed but that you wish to maintain.

Garbage mattes of different modes cannot be mixed within a single tool. A Matte Control node is often used after a Keyer node to add a garbage matte with the opposite effect of the matte applied to the keyer.

Enabling Invert inverts the garbage matte before it is combined with the source alpha.

Post-Multiply Image

Select this option to cause the keyer to multiply the color channels of the image against the alpha channel it creates for the image. This option is usually enabled and is on by default.

Deselect this checkbox and the image can no longer be considered premultiplied for purposes of merging it with other images. Use the Subtractive option of the Merge node instead of the Additive option.

For more information on these Merge node settings, see *Chapter 95, "Composite Nodes,"* in the DaVinci Resolve Reference Manual, or Chapter 35 in the Fusion Reference Manual.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Matte nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

How to Key with Primatte

You begin keying with Primatte by connecting the blue- or green-screen shot to the orange foreground input on the Primatte node and the background shot for the composite into the green background input. Once the connections are made, there are four main steps to using the Primatte:

- 1 Select Background Color.
- 2 Clean the Background Noise.
- 3 Clean the Foreground Noise.
- 4 Remove Spill.

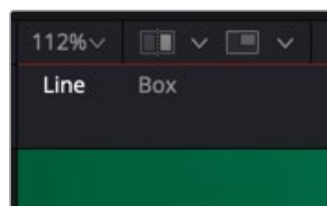
Selecting Background Color

- In the Inspector, click the Select Background Color button.
- Position the mouse pointer over the blue/green-screen area in the viewer, somewhere near the foreground subject.
- Drag over the background color.

Primatte averages the pixels to get a single color. Sometimes Primatte works best when only a single pixel is sampled instead of a range of pixels.

Should you have difficulties with your keying, try the Select Background Color operation again with a single dark screen pixel or single light screen pixel.

Instead, If you want to make a rectangular selection, use the Box button in the top left-hand corner of the viewer. The Median button is the same as Line selection, except that each point sampled is the result of a 3 x 3 region based on where you click and then apply a median filter. This can potentially reduce any noisy pixels.



Primatte viewer buttons

If the foreground image has a shadow in it that you want to keep in the composite, do not select any of the dark screen pixels in the shadow. This keeps the shadow with the rest of the foreground image.

Clean Background Noise

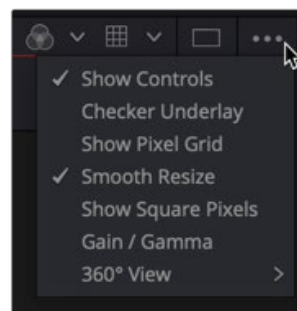
If there are any white or light gray regions in the dark screen area, this is referred to as “noise.” Technically, it is varying shades of the screen color that did not get picked up on the first sample and should be removed. You remove background noise using the Clean Background Noise button.

- 1 From the View Mode menu in the Inspector, select Black.
- 2 Above the viewer, click the Alpha Channel/RGB button.

The image displayed changes to a black and white “matte” view of the image.

- 3 Click the Clean Background Noise button.
 - 4 Drag the mouse pointer through these white or light gray regions that should be pure black.
- Primatte processes the selection and eliminates the noise.
- 5 Repeat this procedure as often as necessary to clear the noise from the background areas.

Selecting Gain/Gamma from the viewer’s Options menu to increase the brightness or gamma allows you to see noise that would otherwise be invisible.



Primatte viewer Options menu

You do not need to remove every single white pixel to get good results. Most pixels displayed as a dark color close to black in a key image are considered transparent and virtually allow the background to be the final output in that area. Consequently, there is no need to eliminate all noise in the screen portions of the image. In particular, if an attempt is made to remove noise around the foreground subject meticulously, a smooth composite image is often difficult to generate.

TIP: When clearing noise from around loose, flying hair or any background/foreground transitional area, be careful not to select any of the areas near the edge of the hair. Leave a little noise around the hair as this can be cleaned up later using the Fine Tuning tools.

Clean Foreground Noise

If there are dark regions in the middle of the mostly white foreground subject, the key is not 100% in those areas. Using Clean Foreground Noise can make the matte as white as possible.

- 1 Keep the View Mode menu set to Black and the viewer set to the Alpha Channel.
- 2 Click the Clean Foreground Noise button.
- 3 Drag the mouse pointer through these dark pixels in the foreground that should be pure white.
Primatte processes the selection and eliminates the noise.
- 4 Repeat this procedure as often as necessary to clear the noise from the foreground areas.
- 5 If enabled, disable Gain/Gamma from the viewer's Options menu to return to a regular viewer.

Removing Spill

The first three sections created a clean matte. At this point, the foreground can be composited onto any background image. However, if there is color spill on the foreground subject, a final operation is necessary to remove that screen spill for a more natural-looking composite.

- 1 From the View Mode menu, select Composite.
- 2 Above the viewer, click the Alpha/RGB toggle button to see the RGB image.

There are two ways in Primatte to remove the spill color:

Spill Sponge

The quickest method is to select the Spill Sponge button and then sample the spill areas away. Additional spill removal can be done using the tools under the Fine Tuning tab or by using the Spill(-) button.

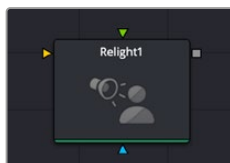
Fine Tuning Tab

To use the Fine Tuning tab for spill, first scrub over the spill color in the viewer. For most images, adjusting the Spill slider is all that is required to remove any remaining spill.

NOTE: When using the slider in the Fine Tuning tab to remove spill, spill color replacement is replaced based on the setting of the Spill Replacement options.

You can use the other two sliders in the same way for different key adjustments. The Detail slider controls the matte softness for the color that is closest to the background color. For example, you can recover lost rarefied smoke in the foreground by selecting the Fine Tuning mode, clicking on the area of the image where the smoke starts to disappear and moving the Detail slider to the left. The Transparency slider controls the matte softness for the color that is closest to the foreground color. For example, if you have thick and opaque smoke in the foreground, you can make it semitransparent by moving the Transparency slider to the right after selecting the pixels in the Fine Tuning mode.

Relight [RLt]



The Relight node

Relight Node Introduction

Relight allows you to add light sources to your scene, realistically illuminating the objects within it. Unlike a simple shape node whose polygon is fixed on screen, Relight analyzes the shapes of people and objects to create a Surface Map and uses that map to reflect light appropriately. When dialed-in properly, this effect can provide the illusion of light following the curves of an object, reflecting off highlights and disappearing into shadows.

Relight is useful in situations when you wished you had another light in the scene, want to accentuate an existing light to match another shot, or want to increase the realism of day-for-night grading.

NOTE: The Relight effect creates the illusion of depth with its analysis of the surfaces in a scene, but it does not calculate 3D relationships between objects in the scene. Shapes cannot be made to cast shadows, and depth information cannot be produced.

For more information on Relight, see *Chapter 162, “Resolve FX Refine.”*

Inputs

The Relight tool has three inputs.

Input: The orange input is used for the primary footage you wish to relight.

Effect Mask: The optional blue effect mask input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the filter to only those pixels within the mask. An effects mask is applied to the tool after the tool is processed.

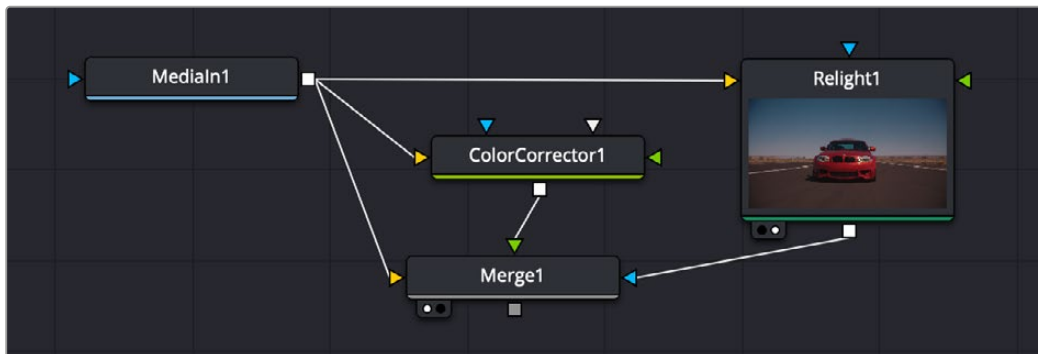
Normals: The green input is used to connect an externally created normal maps, instead of the one generated internally by the Relight tool.

Basic Node Setup

In order to use the Relight tool, you need an additional node structure to build around it. In addition to the standard Medialn node, you will need a Merge node, and another effect like the ColorCorrector or Brightness/Contrast tools.

- **Medialn:** The output from the Medialn node goes to the orange inputs of the Relight, ColorCorrector (or other effect node), and the Merge node.
- **ColorCorrector (effect):** Receives the input from the Medialn node, and its output connects to the green foreground input of the Merge node.

- **Relight:** Receives the input from the MediaIn node, and its output connects to the blue effect mask input on the Merge node. Set this node to appear in the viewer of your choice.
- **Merge:** Receives the background input from the MediaIn node, receives the green foreground input from the ColorCorrector, and the blue effect mask input from Relight. Its output goes to the rest of the node tree or media out. Set this node to appear in the viewer of your choice.



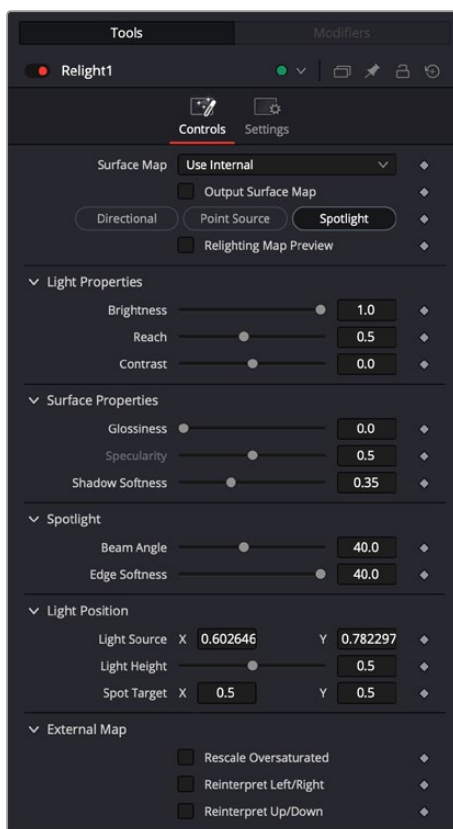
The node tree showing the surrounding node structure for the Relight tool. Note that the ColorCorrector can be replaced by any other preferred effect.

With the above node structure in place, you should see something similar to the setup below. On Viewer 1 is the final relight effect that gets passed onto the rest of the node tree. On Viewer 2 you have the relight interface that allows you to drag the lighting position around the frame. Depending on your Relighting Map Preview option, this will either be a black and white embossed image, or the RGB image with a semi-transparent background.



The Merge node in Viewer 1 (left), and the Relight node in Viewer 2 (right); both sets of on-screen controls are active.

Inspector



The Relight controls

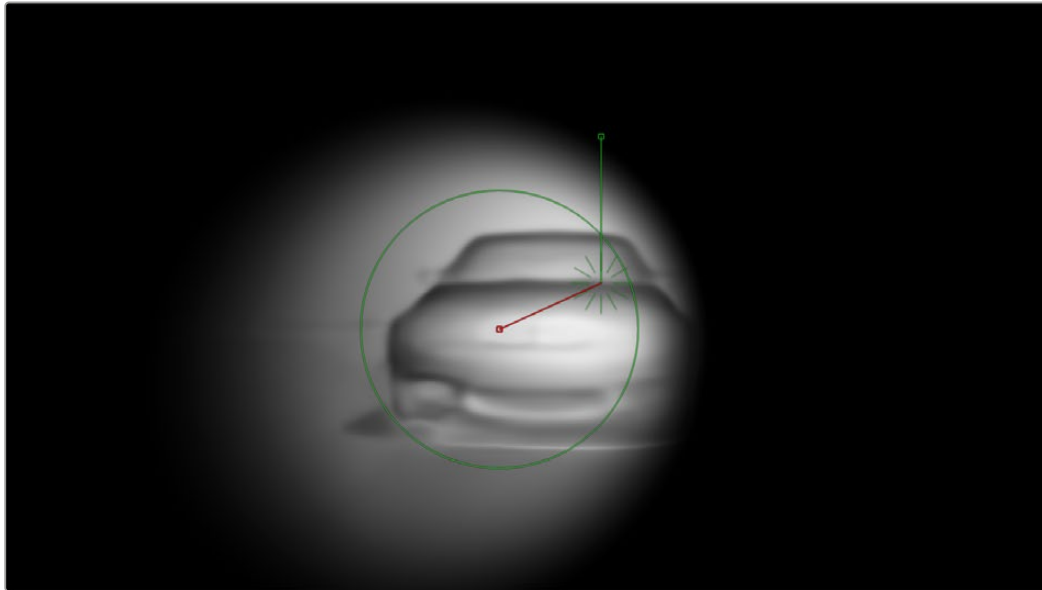
Controls

The Controls let you adjust all the parameters of the Relight tool. There are additional on-screen controls in the viewer as well, depending on the type of Relight effect you choose.

For more information on using Relight controls, see *Chapter 162, “Resolve FX Refine.”*

- **Surface Map:** This chooses which mode you will use to create the Surface Map.
 - Use Internal:** Generates the Surface Map internally to the tool.
 - Use Normals Input:** Receives an externally created Surface Map from another node. Connect that map to the green input of the Relight node.
- **Output Surface Map:** Checking this box causes this copy of the effect to output a multi-colored map of the input frame. This Surface Map can then be fed into a second Relight node where the light sources are placed and adjusted.
- **Directional / Point Source / Spotlight:**
 - Directional:** Creates a light source that emits from a specific direction at an infinite distance.
 - Point Source:** Creates a light source that emits from a central point.
 - Spotlight:** Is also a point source but it is directed into a cone shape.

- **Relighting Map Preview:** Checking this box reveals the Alpha channel of the effect of this light. Unchecking this box shows the current image but with the light mask in the Alpha channel. Brighter areas of the Map Preview show where the Relight is affecting the image.



The image of a car with the Map Preview box checked. Brighter areas show where and how the Relight tool will affect the image.

Light Properties

- **Brightness:** This controls the strength of any grading that is applied to the scene. It does not add brightness changes on its own but amplifies the changes you make in the grading tools.
- **Reach:** This controls how much reflections will dim with distance from the light source.
- **Contrast:** This determines how harsh or gentle the gradient is between light and dark.

Surface Properties

- **Glossiness:** Gives the illusion of a metallic sheen to reflective surfaces.
- **Specularity:** When there is any glossiness added, this controls how metallic/shiny that glossiness is.
- **Shadow Softness:** Controls how much the light penetrates into shadows due to ambient reflections, beyond where it would reach otherwise.

Light Direction (Directional only)

- **Azimuth/Elevation XY :** Controls the direction the light is coming from.

Spotlight (Spotlight only)

- **Beam Angle:** Sets how broad the cone of light is. This control is also available graphically in the viewer.
- **Edge Softness:** Controls the softness of the edges of the light cone/beam. Light Position (Point Source and Spotlight).

Light Position (Point and Spotlight only)

- These controls are for the the position of the light emitter for Point and Spotlight types. They can be adjusted here directly, but they are adjusted graphically using the Viewer controls more easily.

External Map

- **Rescale Oversaturated:** Different applications use different scalings of the Surface Map. A common format looks like the Relight plugin's internal ones but with the colors intensely saturated and can be dark or black in some areas. If your imported surface map has this appearance, check this box to re-interpret it.
- **Reinterpret Left/Right:** Check this box to swap the direction of your external surface map, if your map surfaces are appearing concave instead of convex.
- **Reinterpret Up/Down:** Check this box to swap the direction of your external surface map, if your map surfaces are appearing lit from below when the light is above them.

Ultra Keyer [UKy]



The Ultra Keyer node

Ultra Keyer Node Introduction

Like the newer Delta Keyer, the Ultra Keyer node has two keyers built in to it: a pre-matte keyer acts as a garbage matte creator and the color difference keyer that extracts fine detail and transparency. Generally, you start with the Delta Keyer as your first keyer of choice. If you do not get good results, try Primatte if you are using Fusion Studio. A good third choice is to try the Ultra Keyer.

Inputs

The Ultra Keyer node includes four inputs in the Node Editor.

Input: The orange input accepts a 2D image that contains the color you want to be keyed for transparency.

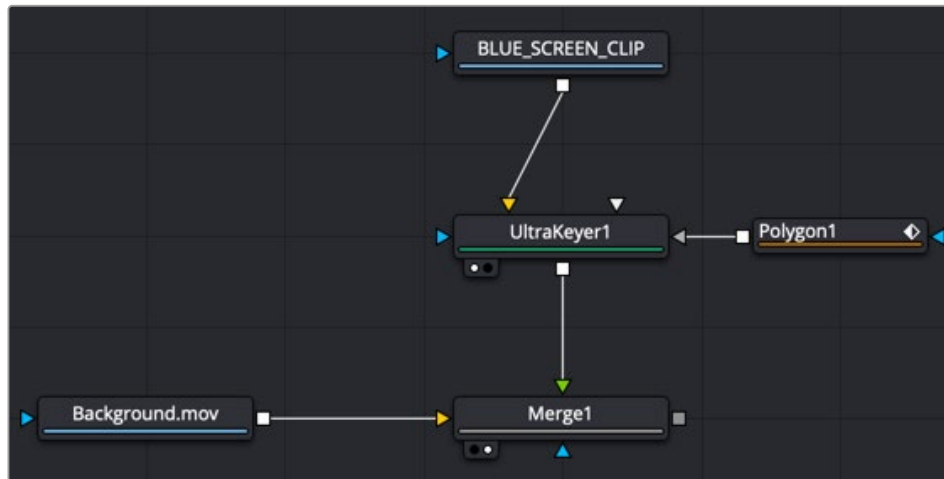
Garbage Matte: The gray garbage matte input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps masks. Connecting a mask to this input causes areas of the image that fall within the matte to be made transparent. The garbage matte is applied directly to the alpha channel of the image.

Solid Matte: The white solid matte input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps masks. Connecting a mask to this input causes areas of the image that fall within the matte to be fully opaque.

Effect Mask: The optional blue input expects a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps masks. Connecting a mask to this input limits the pixels where the keying occurs. An effects mask is applied to the tool after the tool is processed.

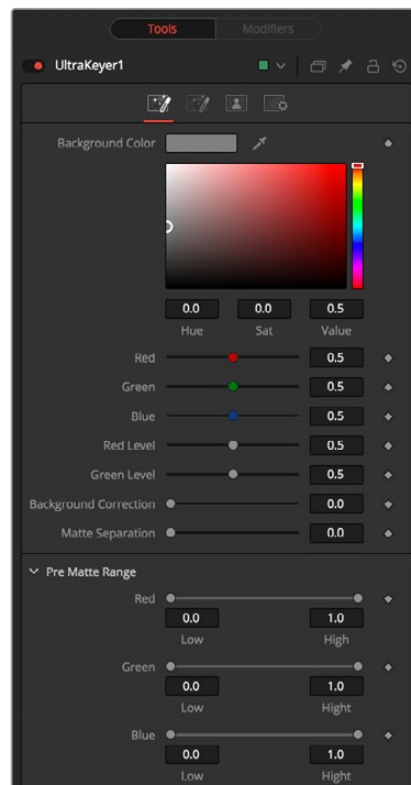
Basic Node Setup

A single keyer can rarely get perfect results because most green- or blue-screen shots have problems the keyer is not made to handle. Keyers often need the help of garbage mattes or solid mattes created with a Polygon or B-Spline node. Shots can also require more than just one keyer to achieve perfect results. Below, the Ultra Keyer node has the blue-screen content connected to the orange input. The result is an image with alpha that can then be connected into the foreground of a Merge node.



An Ultra Keyer node combined with polygon matte as a garbage matte and connected to the foreground of a Merge

Inspector



The Ultra Keyer Pre-Matte tab

Pre-Matte Tab

The Pre-Matte tab is where most keying begins. It is used to select the screen color and smooth out the color of the screen.

Background Color

The Background Color is used to select the color of the blue or green screen of the images. It is good practice to select the screen color close to the subject to be separated from the screen background.

Red Level, Green Level, Blue Level

These color sliders tune the level of the difference channels to help separate the color. When the background color is green, Red and Blue level options are provided. When the background color is blue, Red and Green level options are provided.

Background Correction

Depending on the background color selected above, the keyer iteratively merges the pre-keyed image over either a blue or green background before processing it further.

In some instances, this leads to better, more subtle edges.

Matte Separation

Matte Separation performs a pre-process on the image to help separate the foreground from the background before color selection. Generally, increase this control while viewing the alpha to eliminate the bulk of the background, but stop just before it starts cutting holes in the subject or eroding fine detail on the edges of the matte.

Pre-Matte Range

These R,G,B, and Luminance range controls update automatically to represent the current color selection. Colors are selected by selecting the Ultra Keyer node's tile in the node tree and dragging the Eyedropper into the viewer to select the colors to be used to create the matte. These range controls can be used to tweak the selection slightly, although selecting colors in the viewer is all that is required.

Lock Color Picking

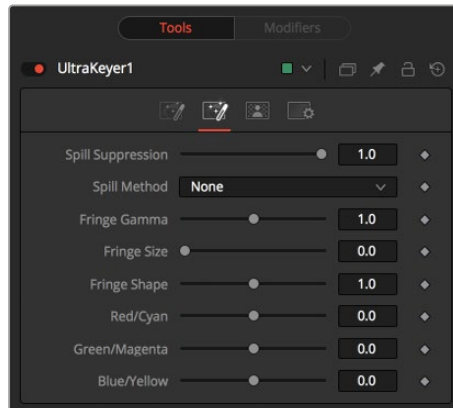
This checkbox prevents accidentally selecting more colors from the view. It is a good idea to activate this checkbox once the color selection is made for the matte. All other controls in the node remain editable.

Pre Matte Size

The Pre Matte Size control can be used to soften the general area around the keyed image. This is used to close holes in the matte often caused by spill in semitransparent areas of the subject. This can cause a small halo around the subject, which can be removed using the Matte Contract tools found later in the tool.

Reset Pre Matte Ranges

This discards all color selection by resetting the ranges but maintains all other slider and control values.



The Ultra Keyer Image tab

Image Tab

The Image tab handles the majority of spill suppression in the Ultra Keyer. Spill suppression is a form of color correction that attempts to remove the screen color from the fringe of the matte.

Spill is the transmission of the screen color through the semitransparent areas of the alpha channel. In the case of blue- or green-screen keying, this usually causes the color of the background to become apparent in the edges of the foreground subject.

Spill Suppression

When this slider is set to 0, no spill suppression is applied to the image. Increasing the slider increases the strength of the spill method.

Spill Method

This selects the strength of the algorithm used to apply spill suppression to the image.

- **None:** None is selected when no spill suppression is required.
- **Rare:** This removes very little of the spill color and is the lightest of all methods.
- **Medium:** This works best for green screens.
- **Well Done:** This works best for blue screens.
- **Burnt:** This works best for blue screens. Use this mode only for very troublesome shots.

Fringe Gamma

This control can be used to adjust the brightness of the fringe or halo that surrounds the keyed image.

Fringe Size

This expands and contracts the size of the fringe or halo surrounding the keyed image.

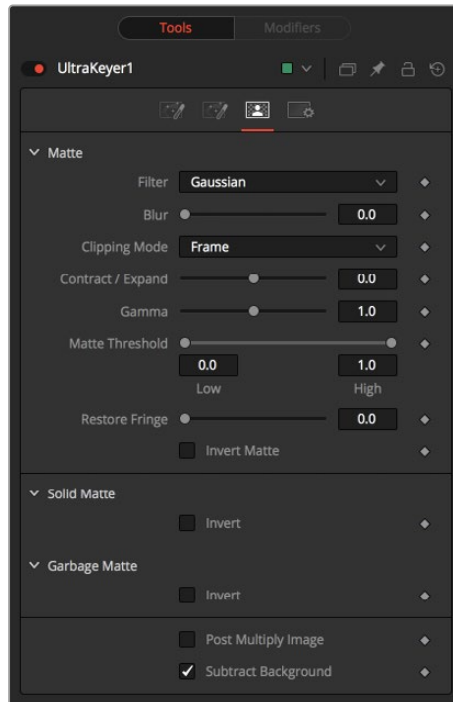
Fringe Shape

Fringe Shape presses the fringe toward the external edge of the image or pulls it toward the inner edge of the fringe. Its effect is most noticeable while the Fringe Size value is large.

Cyan/Red, Magenta/Green, and Yellow/Blue

Use these three controls to color correct the fringe of the image.

This is useful for correcting semitransparent pixels that still contain color from the original background to match the new background.



The Ultra Keyer Matte tab

Matte Tab

The Matte tab refines the alpha of the key, combined with any solid and garbage masks connected to the node. When using the Matte tab, set the viewer to display the alpha channel of the Delta Keyer's final output.

Filter

This control selects the filtering algorithm used when applying blur to the matte.

- **Box:** This is the fastest method but at reduced quality. Box is best suited for minimal amounts of blur.
- **Bartlett:** Otherwise known as a Pyramid filter, Bartlett makes a good compromise between speed and quality.
- **Multi-Box:** When selecting this filter, the Num Passes slider appears and lets you control the quality. At 1 and 2 passes, results are identical to Box and Bartlett, respectively. At 4 passes and above, results are usually as good as Gaussian, in less time and with no edge “ringing.”
- **Gaussian:** The Gaussian filter uses a true Gaussian approximation and gives excellent results, but it is a little slower than the other filters. In some cases, it can produce an extremely slight edge “ringing” on floating-point pixels.

Blur

Matte Blur blurs the edge of the matte based on the Filter menu setting. A value of zero results in a sharp, cutout-like hard edge. The higher the value, the more blur applied to the matte.

Clipping Mode

This option determines how edges are handled when performing domain of definition rendering. This is profoundly important when blurring the matte, which may require samples from portions of the image outside the current domain.

- **Frame:** The default option is Frame, which automatically sets the node's domain of definition to use the full frame of the image, effectively ignoring the current domain of definition. If the upstream DoD is smaller than the frame, the remaining areas in the frame are treated as black/transparent.
- **Domain:** Setting this option to Domain respects the upstream domain of definition when applying the node's effect. This can have adverse clipping effects in situations where the node employs a large filter.
- **None:** Setting this option to None does not perform any source image clipping at all. This means that any data required to process the node's effect that would normally be outside the upstream DoD is treated as black/transparent.

Contract/Expand

This slider shrinks or grows the semitransparent areas of the matte. Values above 0.0 expand the matte, while values below 0.0 contract it.

This control is usually used in conjunction with the Matte Blur to take the hard edge of a matte and reduce fringing. Since this control affects only semitransparent areas, it has no effect on a matte's hard edge.

Gamma

Matte Gamma raises or lowers the values of the matte in the semitransparent areas. Higher values cause the gray areas to become more opaque, and lower values cause the gray areas to become more transparent. Completely black or white regions of the matte remain unaffected.

Since this control affects only semitransparent areas, it will have no effect on a matte's hard edge.

Threshold

This range slider sets the lower threshold using the handle on the left and sets the upper threshold using the handle on the right.

Any value below the lower threshold setting becomes black or transparent in the matte.

Any value above the upper threshold setting becomes white or opaque in the matte. All values within the range maintain their relative transparency values.

This control is often used to reject salt and pepper noise in the matte.

Restore Fringe

This restores the edge of the matte around the keyed subject. Often when keying, the edge of the subject where you have hair is clipped out. Restore Fringe brings back that edge while keeping the matte solid.

Invert Matte

When this checkbox is selected, the alpha channel created by the keyer is inverted, causing all transparent areas to be opaque and all opaque areas to be transparent.

Solid Matte

Solid mattes are mask nodes or images connected to the solid matte input on the node. The solid matte is applied directly to the alpha channel of the image. Generally, solid mattes are used to hold out keying in areas you want to remain opaque, such as someone with blue eyes against a blue screen.

Enabling Invert inverts the solid matte before it is combined with the source alpha.

Garbage Matte

Garbage mattes are mask nodes or images connected to the garbage matte input on the node. The garbage matte is applied directly to the alpha channel of the image. Generally, garbage mattes are used to remove unwanted elements that cannot be keyed, such as microphones and booms. They are also used to fill in areas that contain the color being keyed but that you wish to maintain.

Garbage mattes of different modes cannot be mixed within a single tool. A Matte Control node is often used after a Keyer node to add a garbage matte with the opposite effect of the matte applied to the keyer.

Enabling Invert inverts the garbage matte before it is combined with the source alpha.

Post-Multiply Image

Select this option to cause the keyer to multiply the color channels of the image against the alpha channel it creates for the image. This option is usually enabled and is on by default.

Deselect this checkbox and the image can no longer be considered premultiplied for purposes of merging it with other images. Use the Subtractive option of the Merge node instead of the Additive option.

For more information on these Merge node settings, see *Chapter 95, "Composite Nodes,"* in the DaVinci Resolve Reference Manual, or Chapter 35 in the Fusion Reference Manual.

Subtract Background

This option color corrects the edges when the screen color is removed and anti-aliased to a black background. By enabling this option, the edges potentially become darker. Disabling this option allows you to pass on the color of the screen to use in other processes down the line.

Common Controls

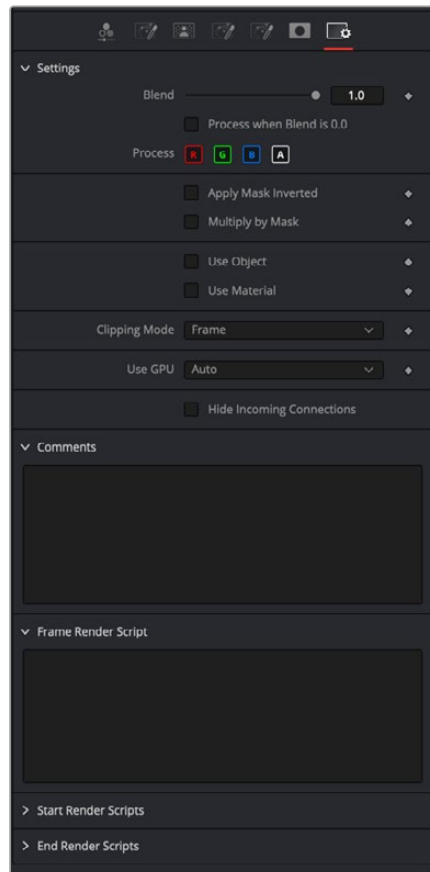
Settings Tab

The Settings tab in the Inspector is also duplicated in other matte nodes. These common controls are described in detail in the following "The Common Controls" section.

The Common Controls

Nodes that handle matte operations share a number of identical controls in the Inspector. This section describes controls that are common among matte nodes.

Inspector



Common Matte settings Inspector

Settings Tab

The Settings tab in the Inspector can be found on every tool in the Matte category. The controls are consistent and work the same way for each tool.

Blend

The Blend control is used to blend between the tool's original image input and the tool's final modified output image. When the blend value is 0.0, the outgoing image is identical to the incoming image. Usually, this causes the tool to skip processing entirely, copying the input straight to the output.

Process When Blend Is 0.0

The tool is processed even when the input value is zero. This can be useful if this node is scripted to trigger another task, but the value is set to 0.0.

Red/Green/Blue/Alpha Channel Selector

These four buttons are used to limit the effect of the tool to specified color channels. This filter is often applied after the tool has been processed.

For example, if the Red button on a Blur tool is deselected, the blur is first applied to the image, and then the red channel from the original input is copied back over the red channel of the result.

There are some exceptions, such as tools where deselecting these channels causes the tool to skip processing that channel entirely. In that case, there are a set of RGBA buttons on the Controls tab in the tool. The buttons in the Settings and the Controls tabs are identical.

Apply Mask Inverted

Enabling the Apply Mask Inverted option inverts the complete mask channel for the tool. The mask channel is the combined result of all masks connected to or generated in a node.

Multiply by Mask

Selecting this option causes the RGB values of the masked image to be multiplied by the mask channel's values. This causes all pixels of the image not included in the mask (i.e., set to 0) to become black/transparent.

Use Object/Use Material (Checkboxes)

Some 3D software can render to file formats that support additional channels. Notably, the EXR file format supports Object ID and Material ID channels, which can be used as a mask for the effect. These checkboxes determine whether the channels are used, if present. The specific Material ID or Object ID affected is chosen using the next set of controls.

Correct Edges

This checkbox appears only when the Use Object or Use Material checkboxes are selected. It toggles the method used to deal with overlapping edges of objects in a multi-object image. When enabled, the Coverage and Background Color channels are used to separate and improve the effect around the edge of the object. If this option disabled (or no Coverage or Background Color channels are available), aliasing may occur on the edge of the mask.

For more information on the Coverage and Background Color channels, see *Chapter 78, "Understanding Image Channels,"* in the DaVinci Resolve Reference Manual, or Chapter 18 in the Fusion Reference Manual.

Object ID/Material ID (Sliders)

Use these sliders to select which ID is used to create a mask from the object or material channels of an image. Use the Sample button in the same way as the Color Picker: to grab IDs from the image displayed in the view. The image or sequence must have been rendered from a 3D software package with those channels included.

Clipping Mode

This option determines how edges are handled when performing domain of definition rendering. This is profoundly important when blurring the matte, which may require samples from portions of the image outside the current domain.

- **Frame:** The default option is Frame, which automatically sets the node's domain of definition to use the full frame of the image, effectively ignoring the current domain of definition. If the upstream DoD is smaller than the frame, the remaining area in the frame is treated as black/transparent.
- **Domain:** Setting this option to Domain respects the upstream domain of definition when applying the node's effect. This can have adverse clipping effects in situations where the node employs a large filter.
- **None:** Setting this option to None does not perform any source image clipping at all. This means that any data required to process the node's effect that would normally be outside the upstream DoD is treated as black/transparent.

Use GPU

The Use GPU menu has three settings. Setting the menu to Disable turns off GPU hardware-accelerated rendering. Enabled uses the GPU hardware for rendering the node. Auto uses a capable GPU if one is available, and falls back to software rendering when a capable GPU is not available.

Hide Incoming Connections

Enabling this checkbox can hide connection lines from incoming nodes, making a node tree appear cleaner and easier to read. When enabled, empty fields for each input on a node are displayed in the Inspector. Dragging a connected node from the node tree into the field hides that incoming connection line as long as the node is not selected in the node tree. When the node is selected in the node tree, the line reappears.

Comments

The Comments field is used to add notes to a tool. Click in the empty field and type the text. When a note is added to a tool, a small red square appears in the lower-left corner of the node when the full tile is displayed, or a small text bubble icon appears on the right when nodes are collapsed. To see the note in the Node Editor, hold the mouse pointer over the node to display the tooltip.

Scripts

Three Scripting fields are available on every tool in Fusion from the Settings tab. They each contain edit boxes used to add scripts that process when the tool is rendering. For more details on scripting nodes, please consult the Fusion scripting documentation.